

Klein Model for Hyperbolic Neural Networks

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Abstract

Hyperbolic neural networks (HNNs) have been proved effective in modeling complex data structures. However, previous works mainly focused on the Poincaré ball model and the hyperboloid model as coordinate representations of the hyperbolic space, often neglecting the Klein model. Despite this, the Klein model offers its distinct advantages thanks to its straight-line geodesics, which facilitates the well-known Einstein midpoint construction, previously leveraged to accompany HNNs in other models. In this work, we introduce a framework for hyperbolic neural networks based on the Klein model. We provide a detailed formulation for representing useful operations using the Klein model. We further study the Klein linear layer and prove that the “tangent space construction” of the scalar multiplication and parallel transport are exactly the Einstein scalar multiplication and the Einstein addition, analogous to the Möbius operations used in the Poincaré ball model. We show numerically that Klein HNNs perform on par with HNNs using the Poincaré ball model, providing a third option for HNN that works as a building block for more complicated architectures.

Keywords: Hyperbolic neural network, Klein model, Einstein gyrovector space

1. Introduction

Hyperbolic spaces have shown considerable promise in embedding complex networks (Krioukov et al., 2010), trees (Wilson et al., 2014; Sonthalia and Gilbert, 2020) and hierarchical datasets (Nickel and Kiela, 2018). To leverage the inherent geometric structures within these data types in learning neural representations, hyperbolic neural networks (HNNs) were introduced, initially by Ganea et al. (2018a) and later expanded upon by many recent works (Peng et al., 2021). To define neural operations in hyperbolic spaces, HNNs typically utilize a model of hyperbolic geometry, where points are represented by Euclidean coordinates. Most of the literature focuses on either the Poincaré ball model or the hyperboloid model. Indeed, Ganea et al. (2018a); Shimizu et al. (2021) presented a set of operations for implementing HNNs using the Poincaré ball model, which often enjoys a simple mathematical description using Möbius operations. On the other hand, the hyperboloid model proves

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to be numerically more stable than the Poincaré ball model (Nickel and Kiela, 2018; Qu and Zou, 2022; Mishne et al., 2023) and facilitates neural operations fully in the hyperbolic space without using tangent spaces (Chen et al., 2022).

Despite the success of HNNs using the Poincaré ball and the hyperboloid models, HNNs using other models remain largely unexplored. This raises problems when important operations are defined only in a certain model. Indeed, in many works (Gulcehre et al., 2019; Zhu et al., 2020; Khrulkov et al., 2020; Zhang and Gao, 2021; Tai et al., 2021; Song et al., 2022; Fu et al., 2024; Skrodzki et al., 2024; Li et al., 2024a), in order to use the Einstein midpoint to perform aggregation, a mapping between the Poincaré ball/hyperboloid model and the Klein model has to be implemented whenever aggregation is needed. This causes unnecessary computational complexity. The Klein model also enjoys other crucial properties such as straight-line geodesics. However, to the best of our knowledge, no prior works have discussed compact neural operations using the Klein model, that is, “pure Klein” HNNs. Moreover, recent software packages for HNNs lack implementation in the Klein model (van Spengler et al., 2023).

In this paper, we derive compact formulas for key operations in the Klein model. Analogous to the Möbius operations in the Poincaré ball model, the Klein model facilitates a set of operations known as Einstein scalar multiplication and Einstein addition (Ungar, 2009, 2012). For the Poincaré ball model, Ganea et al. (2018a) demonstrated that the Möbius operations are exactly the exponential maps of “linear” operations (as used in multilayer perceptrons for obtaining pre-activations) in tangent spaces. However, what remains unclear is the precise relationship between Einstein operations and similar “tangent space constructions” in the Klein model. In our paper, we prove that they are equivalent operations. Thanks to this interpretation, we build Klein HNNs based on these Einstein operations. Through experiments on well-known graph datasets with hierarchical structures, we show that the performance of Klein HNNs is on par with both Poincaré HNNs and hyperboloid HNNs. Furthermore, Klein HNNs are efficient to implement.

2. Related Works

Hyperbolic neural networks The first HNN was introduced relatively recently by Ganea et al. (2018a), who utilized the Poincaré ball model to develop fundamental neural operations including linear layers and recurrent layers. Importantly, they demonstrated that the Möbius operations correspond to the tangent space operations that one would intuitively define for neural operations in the Poincaré ball model. Later, Shimizu et al. (2021) extended upon their work and introduced more operations. Another line of research employed the hyperboloid model, favored for its numerical stability (Gulcehre et al., 2019; Chen et al., 2022). More complex hyperbolic neural operations have since been designed for a variety of applications, particularly for graph data (Chami et al., 2019; Liu et al., 2019; Mathieu et al., 2019; Dai et al., 2021; Zhang et al., 2021; Yang et al., 2022, 2024b) and image data (Khrulkov et al., 2020; Atigh et al., 2022; Ermolov et al., 2022; Desai et al., 2023; Yang et al., 2024c). Additionally, there have been studies that focused on the numerical stability (Mishne et al., 2023) and robustness (Li et al., 2024b) of HNNs. Despite these advances, most models continue to rely on either the Poincaré ball or the hyperboloid model, with other models of hyperbolic geometry largely overlooked.

The Klein model in use Recently, the Klein model has been proved effective in representing various types of data, including hierarchical graphs (McDonald and He, 2020; Yang et al., 2024a), protein sequences (Ali et al., 2024), minimal spanning trees (García-Castellanos et al., 2024), and scene images (Bi et al., 2017). A key advantage of the Klein model is that its geodesics are represented as straight lines, which facilitates intuitive constructions of Voronoi diagrams (Nielsen and Nock, 2010), Delaunay graphs (Medbouhi et al., 2024) and SVM decision boundaries (Cho et al., 2019). Celińska-Kopczyńska and Kopczyński (2024) studied the numerical precision of hyperbolic models including the Klein model. Moreover, properties of the Klein model are also used to assist in proofs for other hyperbolic models (He et al., 2024). However, the above works did not explore neural operations in the Klein model.

Although early works (Taherian, 2010; Rostamzadeh and Taherian, 2014) studied the algebraic structures and trigonometry of the Klein model, they did not consider constructing neural network layers. While Ungar (2009, 2012); Kim and Lawson (2013) summarized the gyrogroup properties of the Einstein operations in connection with the Klein model, they did not mention the properties that we introduce in this paper.

3. The Klein Model of Hyperbolic Geometry

The n -dimensional hyperbolic space, denoted as $\mathbb{H}^n$, is a simply connected n -dimensional Riemannian manifold with constant negative curvature. To represent points in $\mathbb{H}^n$, there are several isometric models including the Poincaré ball model $\mathbb{B}^n$ and the hyperboloid (Lorentz) model $\mathbb{L}^n$, both of which are commonly used in the HNN literature. The Klein model (also known as Beltrami-Klein), denoted as $\mathbb{K}^n$, is another notable representation. In our paper, we consider $\mathbb{H}^n$ with a curvature -1 since this is used in most of the HNN literature.

Recent works have derived compact formulas for basic operations in $\mathbb{B}^n$ and $\mathbb{L}^n$ (Ganea et al., 2018a; Chami et al., 2019). Our derivation will use these formulas to avoid complicated calculations of, e.g., connections. In this section, we introduce the Klein model and describe the isometric mappings between the Klein model and the other two models. A detailed review of $\mathbb{B}^n$ and $\mathbb{L}^n$ is provided in Appendix A.

The n -dimensional Klein model with a curvature -1 is represented as $\mathbb{K}^n = \{\mathbf{x} \in \mathbb{R}^n, \|\mathbf{x}\| < 1\}$, with the Riemannian metric tensor $g_{\mathbf{x}}^{\mathcal{K}} = g^{\mathcal{K}}(\mathbf{x})$ expressed as:

$$g_{ij}^{\mathcal{K}}(\mathbf{x}) = \frac{\delta_{ij}}{1 - \|\mathbf{x}\|^2} + \frac{x_i x_j}{(1 - \|\mathbf{x}\|^2)^2}, \quad (1)$$

where $\|\cdot\|$ is the Euclidean norm. Denoting points in $\mathbb{R}^{n+1}$ as $[x_0, x_1, \dots, x_n]^\top$, the Klein ball can be obtained by mapping $\mathbf{x} \in \mathbb{L}^n$ to the hyperplane $x_0 = 1$, using a gnomonic (central) projection with rays emanating from the origin. Specifically, we review the following formulas for isometric mappings between hyperbolic models. We say two points are “corresponding” if they are related by such an isometric mapping.

Mappings between models We use $\pi_{\mathcal{L} \rightarrow \mathcal{K}}$ to denote the isometric mapping from $\mathbb{L}^n$ to $\mathbb{K}^n$ and $\pi_{\mathcal{K} \rightarrow \mathcal{L}}$ the isometric mapping from $\mathbb{K}^n$ to $\mathbb{L}^n$. Namely, for $\mathbf{x}^{\mathcal{L}} = [x_0^{\mathcal{L}}, x_1^{\mathcal{L}}, \dots, x_n^{\mathcal{L}}]^\top \in$

$\mathbb{L}^n$ and $\mathbf{x}^{\mathcal{K}} = [x_1^{\mathcal{K}}, \dots, x_n^{\mathcal{K}}]^\top \in \mathbb{K}^n$,

$$\pi_{\mathcal{L} \rightarrow \mathcal{K}}(\mathbf{x}^{\mathcal{L}}) = \left[\frac{x_1^{\mathcal{L}}}{x_0^{\mathcal{L}}}, \dots, \frac{x_n^{\mathcal{L}}}{x_0^{\mathcal{L}}} \right]^\top \in \mathbb{K}^n, \quad \pi_{\mathcal{K} \rightarrow \mathcal{L}}(\mathbf{x}^{\mathcal{K}}) = \frac{1}{\sqrt{1 - \|\mathbf{x}^{\mathcal{K}}\|^2}} [1, \mathbf{x}^{\mathcal{K}}]^\top \in \mathbb{L}^n. \quad (2)$$

We use $\pi_{\mathcal{B} \rightarrow \mathcal{K}}$ to denote the isometric mapping from $\mathbb{B}^n$ to $\mathbb{K}^n$ and $\pi_{\mathcal{K} \rightarrow \mathcal{B}}$ the isometric mapping from $\mathbb{K}^n$ to $\mathbb{B}^n$. Namely, for $\mathbf{x}^{\mathcal{B}} \in \mathbb{B}^n$ and $\mathbf{x}^{\mathcal{K}} \in \mathbb{K}^n$,

$$\pi_{\mathcal{B} \rightarrow \mathcal{K}}(\mathbf{x}^{\mathcal{B}}) = \frac{2}{1 + \|\mathbf{x}^{\mathcal{B}}\|^2} \mathbf{x}^{\mathcal{B}} \in \mathbb{K}^n, \quad \pi_{\mathcal{K} \rightarrow \mathcal{B}}(\mathbf{x}^{\mathcal{K}}) = \frac{1}{1 + \sqrt{1 - \|\mathbf{x}^{\mathcal{K}}\|^2}} \mathbf{x}^{\mathcal{K}} \in \mathbb{B}^n. \quad (3)$$

In $\mathbb{K}^n$, the tangent space $\mathcal{T}_{\mathbf{x}^{\mathcal{K}}} \mathbb{K}^n$ at $\mathbf{x}^{\mathcal{K}}$ is represented as $\mathbb{R}^n$ where $\frac{\partial}{\partial x_i^{\mathcal{K}}}$ is represented as $\mathbf{e}_i$, the i -th canonical basis. Similar conventions are adopted for $\mathbb{B}^n$ and $\mathbb{L}^n$. The tangent vectors are related using pushforward maps associated with the above isometric mappings. We say that two vectors are “corresponding” if they are related by such a pushforward map. We recall that a pushforward map of an isometry preserves inner products and parallel transports.

The following lemma presents formulas for obtaining the corresponding tangent vectors between $\mathbb{K}^n$ and other models. The proofs of all the results are presented in Appendix B.

Lemma 1 *Let $\mathbf{x}^{\mathcal{K}} \in \mathbb{K}^n$ and $\mathbf{v}^{\mathcal{K}} \in \mathcal{T}_{\mathbf{x}^{\mathcal{K}}} \mathbb{K}^n$ be a tangent vector at $\mathbf{x}^{\mathcal{K}}$.*

1. *Let $\mathbf{x}^{\mathcal{B}} \in \mathbb{B}^n$ be the corresponding point of $\mathbf{x}^{\mathcal{K}}$ and $\mathbf{v}^{\mathcal{B}} \in \mathcal{T}_{\mathbf{x}^{\mathcal{B}}} \mathbb{B}^n$ be the corresponding tangent vector of $\mathbf{x}^{\mathcal{K}}$. Then*

$$\mathbf{v}^{\mathcal{K}} = \frac{\partial \pi_{\mathcal{B} \rightarrow \mathcal{K}}(\mathbf{x}^{\mathcal{B}})}{\partial \mathbf{x}^{\mathcal{B}}} \mathbf{v}^{\mathcal{B}} = \frac{2}{1 + \|\mathbf{x}^{\mathcal{B}}\|^2} \mathbf{v}^{\mathcal{B}} - \frac{4\mathbf{x}^{\mathcal{B}} \cdot \mathbf{v}^{\mathcal{B}}}{(1 + \|\mathbf{x}^{\mathcal{B}}\|^2)^2} \mathbf{x}^{\mathcal{B}}, \quad (4)$$

$$\begin{aligned} \mathbf{v}^{\mathcal{B}} &= \frac{\partial \pi_{\mathcal{K} \rightarrow \mathcal{B}}(\mathbf{x}^{\mathcal{K}})}{\partial \mathbf{x}^{\mathcal{K}}} \mathbf{v}^{\mathcal{K}} \\ &= \frac{1}{1 + \sqrt{1 - \|\mathbf{x}^{\mathcal{K}}\|^2}} \mathbf{v}^{\mathcal{K}} + \frac{\mathbf{x}^{\mathcal{K}} \cdot \mathbf{v}^{\mathcal{K}}}{\sqrt{1 - \|\mathbf{x}^{\mathcal{K}}\|^2} (1 + \sqrt{1 - \|\mathbf{x}^{\mathcal{K}}\|^2})^2} \mathbf{x}^{\mathcal{K}}. \end{aligned} \quad (5)$$

2. *Let $\mathbf{x}^{\mathcal{L}} \in \mathbb{L}^n$ be the corresponding point of $\mathbf{x}^{\mathcal{K}}$ and $\mathbf{v}^{\mathcal{L}} \in \mathcal{T}_{\mathbf{x}^{\mathcal{L}}} \mathbb{L}^n$ be the corresponding tangent vector of $\mathbf{x}^{\mathcal{K}}$. Then*

$$\mathbf{v}^{\mathcal{K}} = \frac{\partial \pi_{\mathcal{L} \rightarrow \mathcal{K}}(\mathbf{x}^{\mathcal{L}})}{\partial \mathbf{x}^{\mathcal{L}}} \mathbf{v}^{\mathcal{L}} = -\frac{v_t}{x_t^2} \mathbf{x}_s + \frac{1}{x_t} \mathbf{v}_s, \quad (6)$$

$$\mathbf{v}^{\mathcal{L}} = \frac{\partial \pi_{\mathcal{K} \rightarrow \mathcal{L}}(\mathbf{x}^{\mathcal{K}})}{\partial \mathbf{x}^{\mathcal{K}}} \mathbf{v}^{\mathcal{K}} = \left[\begin{array}{c} \frac{\mathbf{x}^{\mathcal{K}} \cdot \mathbf{v}^{\mathcal{K}}}{(1 - \|\mathbf{x}^{\mathcal{K}}\|^2)^{\frac{3}{2}}} \\ \frac{1}{\sqrt{1 - \|\mathbf{x}^{\mathcal{K}}\|^2}} \mathbf{v}^{\mathcal{K}} + \frac{\mathbf{x}^{\mathcal{K}} \cdot \mathbf{v}^{\mathcal{K}}}{(1 - \|\mathbf{x}^{\mathcal{K}}\|^2)^{\frac{3}{2}}} \mathbf{x}^{\mathcal{K}} \end{array} \right]. \quad (7)$$

Distances The induced distance function on the Klein model can be inferred from the one on the hyperboloid model, which we present as the following lemma.

Lemma 2 *Let $\mathbf{x}^{\mathcal{K}}, \mathbf{y}^{\mathcal{K}} \in \mathbb{K}^n$. Their geodesic distance is given by*

$$d_{\mathcal{K}}(\mathbf{x}^{\mathcal{K}}, \mathbf{y}^{\mathcal{K}}) = d_{\mathcal{L}}(\pi_{\mathcal{K} \rightarrow \mathcal{L}}(\mathbf{x}^{\mathcal{K}}), \pi_{\mathcal{K} \rightarrow \mathcal{L}}(\mathbf{y}^{\mathcal{K}})) = \cosh^{-1} \left(\frac{1 - \mathbf{x}^{\mathcal{K}} \cdot \mathbf{y}^{\mathcal{K}}}{\sqrt{1 - \|\mathbf{x}^{\mathcal{K}}\|^2} \sqrt{1 - \|\mathbf{y}^{\mathcal{K}}\|^2}} \right). \quad (8)$$

Unit-speed geodesics The parametric expression of the unit-speed geodesics in the Klein model is given in the following lemma, which is derived based on the fact that an isometric mapping maps the geodesics in $\mathbb{L}^n$ to the geodesics in $\mathbb{K}^n$. We remark that the formula for constant-speed geodesics with specified endpoints in the Klein model is provided in (Nielsen and Nock, 2012).

Lemma 3 Let $\mathbf{x}^\mathcal{K} \in \mathbb{K}^n$ and $\mathbf{v}^\mathcal{K} \in \mathcal{T}_{\mathbf{x}^\mathcal{K}}\mathbb{K}^n$ with $g_{\mathbf{x}^\mathcal{K}}^\mathcal{K}(\mathbf{v}^\mathcal{K}, \mathbf{v}^\mathcal{K}) = 1$. Let $\gamma_{\mathbf{x}^\mathcal{K}, \mathbf{v}^\mathcal{K}}(t)$ denote the unit-speed geodesic in $\mathbb{K}^n$ with $\gamma_{\mathbf{x}^\mathcal{K}, \mathbf{v}^\mathcal{K}}(0) = \mathbf{x}^\mathcal{K}$ and $\dot{\gamma}_{\mathbf{x}^\mathcal{K}, \mathbf{v}^\mathcal{K}}(0) = \mathbf{v}^\mathcal{K}$, then

$$\gamma_{\mathbf{x}^\mathcal{K}, \mathbf{v}^\mathcal{K}}(t) = \mathbf{x}^\mathcal{K} + \frac{\sinh(t)\mathbf{v}^\mathcal{K}}{\cosh(t) + \lambda_{\mathbf{x}^\mathcal{K}}^2(\mathbf{x}^\mathcal{K} \cdot \mathbf{v}^\mathcal{K}) \sinh(t)}, \quad (9)$$

where $\lambda_{\mathbf{x}^\mathcal{K}} := 1/\sqrt{1 - \|\mathbf{x}^\mathcal{K}\|^2}$ is used globally in this paper.

Accordingly, we can write out the exponential maps and logarithmic maps for the Klein model, summarized as the following corollary.

Corollary 4 Given $\mathbf{x}^\mathcal{K} \in \mathbb{K}^n$ and $\mathbf{v}^\mathcal{K} \in \mathcal{T}_{\mathbf{x}^\mathcal{K}}\mathbb{K}^n$, denote $\mathbf{u}^\mathcal{K} = \mathbf{v}^\mathcal{K}/\|\mathbf{v}^\mathcal{K}\|_\mathcal{K}$ with $\|\cdot\|_\mathcal{K}^2 = g_{\mathbf{x}^\mathcal{K}}^\mathcal{K}(\cdot, \cdot)$. Let $\gamma_{\mathbf{x}^\mathcal{K}, \mathbf{u}^\mathcal{K}}(t)$ be the unit-speed geodesic in $\mathbb{K}^n$ with $\gamma_{\mathbf{x}^\mathcal{K}, \mathbf{u}^\mathcal{K}}(0) = \mathbf{x}^\mathcal{K}$ and $\dot{\gamma}_{\mathbf{x}^\mathcal{K}, \mathbf{u}^\mathcal{K}}(0) = \mathbf{u}^\mathcal{K}$, the exponential map $\exp_{\mathbf{x}^\mathcal{K}}^\mathcal{K} : \mathcal{T}_{\mathbf{x}^\mathcal{K}}\mathbb{K}^n \rightarrow \mathbb{K}^n$ is given by

$$\exp_{\mathbf{x}^\mathcal{K}}^\mathcal{K}(\mathbf{v}^\mathcal{K}) = \gamma_{\mathbf{x}^\mathcal{K}, \mathbf{u}^\mathcal{K}}(\|\mathbf{v}^\mathcal{K}\|_\mathcal{K}) = \mathbf{x}^\mathcal{K} + \frac{\sinh(\|\mathbf{v}^\mathcal{K}\|_\mathcal{K}) \frac{\mathbf{v}^\mathcal{K}}{\|\mathbf{v}^\mathcal{K}\|_\mathcal{K}}}{\cosh(\|\mathbf{v}^\mathcal{K}\|_\mathcal{K}) + \frac{\lambda_{\mathbf{x}^\mathcal{K}}^2}{\|\mathbf{v}^\mathcal{K}\|_\mathcal{K}}(\mathbf{x}^\mathcal{K} \cdot \mathbf{v}^\mathcal{K}) \sinh(\|\mathbf{v}^\mathcal{K}\|_\mathcal{K})}. \quad (10)$$

The logarithmic map $\log_{\mathbf{x}^\mathcal{K}}^\mathcal{K} : \mathbb{K}^n \rightarrow \mathcal{T}_{\mathbf{x}^\mathcal{K}}\mathbb{K}^n$ is given by

$$\log_{\mathbf{x}^\mathcal{K}}^\mathcal{K}(\mathbf{y}^\mathcal{K}) = d_\mathcal{K}(\mathbf{x}^\mathcal{K}, \mathbf{y}^\mathcal{K}) \frac{\mathbf{y}^\mathcal{K} - \mathbf{x}^\mathcal{K}}{\|\mathbf{y}^\mathcal{K} - \mathbf{x}^\mathcal{K}\|_\mathcal{K}} = \cosh^{-1} \left(\frac{1 - \mathbf{x}^\mathcal{K} \cdot \mathbf{y}^\mathcal{K}}{\sqrt{1 - \|\mathbf{x}^\mathcal{K}\|^2} \sqrt{1 - \|\mathbf{y}^\mathcal{K}\|^2}} \right) \frac{\mathbf{y}^\mathcal{K} - \mathbf{x}^\mathcal{K}}{\|\mathbf{y}^\mathcal{K} - \mathbf{x}^\mathcal{K}\|_\mathcal{K}}, \quad (11)$$

In particular, it follows that

$$\exp_{\mathbf{o}^\mathcal{K}}^\mathcal{K}(\mathbf{v}^\mathcal{K}) = \tanh(\|\mathbf{v}^\mathcal{K}\|) \frac{\mathbf{v}^\mathcal{K}}{\|\mathbf{v}^\mathcal{K}\|}, \quad \log_{\mathbf{o}^\mathcal{K}}^\mathcal{K}(\mathbf{y}^\mathcal{K}) = \cosh^{-1}(\lambda_{\mathbf{y}^\mathcal{K}}) \frac{\mathbf{y}^\mathcal{K}}{\|\mathbf{y}^\mathcal{K}\|}. \quad (12)$$

Parallel transport The parallel transport $P_{\mathbf{x} \rightarrow \mathbf{y}}$ defines a linear isometry moving tangent vectors along the geodesic from $\mathbf{x}$ to $\mathbf{y}$. Unfortunately, the formula for parallel transport between general points would be complicated in the Klein model. We only present the case where the starting point is $\mathbf{o}^\mathcal{K}$, the hyperbolic origin in $\mathbb{K}^n$, which is sufficient for the use case of hyperbolic neural networks.

Proposition 5 The parallel transport of a tangent vector $\mathbf{v}^\mathcal{K} \in \mathcal{T}_{\mathbf{o}^\mathcal{K}}\mathbb{K}^n$ to the tangent space $\mathcal{T}_{\mathbf{x}^\mathcal{K}}\mathbb{K}^n$ at an arbitrary point $\mathbf{x}^\mathcal{K}$ in the Klein model is

$$P_{\mathbf{o}^\mathcal{K} \rightarrow \mathbf{x}^\mathcal{K}}(\mathbf{v}^\mathcal{K}) = \frac{(\mathbf{x}^\mathcal{K} \cdot \mathbf{v}^\mathcal{K})(\sqrt{1 - \|\mathbf{x}^\mathcal{K}\|^2} - 2)}{1 - \sqrt{1 - \|\mathbf{x}^\mathcal{K}\|^2}} \mathbf{x}^\mathcal{K} + \sqrt{1 - \|\mathbf{x}^\mathcal{K}\|^2} \mathbf{v}^\mathcal{K} \in \mathcal{T}_{\mathbf{x}^\mathcal{K}}\mathbb{K}^n. \quad (13)$$

4. Klein Model for Hyperbolic Neural Networks

To build HNNs in hyperbolic spaces, we need to define operations including weight matrix transformation and bias translation in a way analogous to “ $\mathbf{W}\mathbf{x} + \mathbf{b}$ ”. A natural approach is as follows. First, apply the logarithmic map to project the input to the tangent space at the hyperbolic origin. Then for weight matrix transformation, perform the matrix-vector multiplication in the tangent space; for bias translation, apply the parallel transport of the projection to the tangent space at the input point. Finally, for both operations, use the exponential map to bring the result back to hyperbolic space. For the Poincaré ball model, [Ganea et al. \(2018a\)](#) proved that these “tangent space operations” are equivalent to basic operations in the Möbius gyrovector spaces. In the context of the Klein model, the corresponding gyrovector space is the Einstein gyrovector space. This raises a natural question: are the tangent space operations in the Klein model also equivalent to those in the Einstein gyrovector space? If so, we can find a geometric interpretation for the neural operations in the Klein model and adopt Einstein gyrovector operations to save the tangent space mappings in Klein HNNs.

In Section 4.1, we review the definitions of the Einstein gyrovector spaces. We prove the equivalence of tangent space operations and Einstein gyrovector space operations in Section 4.2. Finally, we derive HNNs in the Klein model in Section 4.3.

4.1. Einstein Gyrogroups and Gyrovector Spaces

Einstein gyrogroup Let $\mathbb{V}$ be a real inner product space and let $\mathbb{V}_c = \{\mathbf{x} \in \mathbb{V} : \|\mathbf{x}\| < c\}$ be an open c -ball of $\mathbb{V}$. Then the Einstein addition of $\mathbf{x}, \mathbf{y} \in \mathbb{V}_c$ is defined as

$$\mathbf{x} \oplus_E \mathbf{y} = \frac{1}{1 + \frac{\mathbf{x} \cdot \mathbf{y}}{c^2}} \left(\mathbf{x} + \frac{1}{\gamma_{\mathbf{x}}} \mathbf{y} + \frac{1}{c^2} \frac{\gamma_{\mathbf{x}}}{1 + \gamma_{\mathbf{x}}} (\mathbf{x} \cdot \mathbf{y}) \mathbf{x} \right), \quad (14)$$

where $\gamma_{\mathbf{x}} = 1/\sqrt{1 - \frac{\|\mathbf{x}\|^2}{c^2}}$ is the Lorentz factor. These designations originate from Einstein’s theory of special relativity: if $\mathbb{V} = \mathbb{R}^3$, Equation (14) represents the relativistic addition of velocities $\mathbf{x}, \mathbf{y}$ in space rather than spacetime, where c is the speed of light.

In the general case where $\mathbf{x}, \mathbf{y}$ are not parallel, $\oplus_E$ is neither commutative nor associative, rendering the corresponding algebraic structure merely a groupoid. As observed by Ungar, however, this groupoid has the interesting property whereby an automorphism $\text{gyr}[\mathbf{x}, \mathbf{y}] : \mathbb{V}_c \rightarrow \mathbb{V}_c$ respecting $\oplus_E$, called gyration, defines the obstruction to commutativity through $\mathbf{x} \oplus_E \mathbf{y} = \text{gyr}[\mathbf{x}, \mathbf{y}] \mathbf{y} \oplus_E \mathbf{x}$ as well as to associativity through $\mathbf{x} \oplus_E (\mathbf{y} \oplus_E \mathbf{z}) = (\mathbf{x} \oplus_E \mathbf{y}) \oplus_E \text{gyr}[\mathbf{x}, \mathbf{y}] \mathbf{z}$. This groupoid structure $(\mathbb{V}_c, \oplus_E)$ is called a gyrocommutative gyrogroup, and describes the Thomas precession in the example of special relativity. We refer to $(\mathbb{V}_c, \oplus_E)$ as Einstein gyrogroup for short, and more details on its properties can be found, e.g., in [Ungar \(2009, sec. 1.2, 2.3\)](#).

Einstein gyrovector space Given an Einstein gyrogroup, we can endow it with another operation called Einstein scalar multiplication: letting $r \in \mathbb{R}$, $\mathbf{x} \in \mathbb{V}_c$, and $\mathbf{x} \neq \mathbf{0}$, it is

$$r \otimes_E \mathbf{x} = c \tanh \left(r \tanh^{-1} \left(\frac{\|\mathbf{x}\|}{c} \right) \right) \frac{\mathbf{x}}{\|\mathbf{x}\|}. \quad (15)$$

This turns out to have associative and distributive properties, and we apply the notation $r \otimes_E \mathbf{x} = \mathbf{x} \otimes_E r$. The resulting structure $(\mathbb{V}_c, \oplus_E, \otimes_E)$ is called Einstein gyrovector space, and see, e.g., Ungar (2009, sec. 3.1, 3.8) for relevant properties. Notably, its inherited inner product is invariant under gyrations, that is, $\text{gyr}[\mathbf{x}, \mathbf{y}] \mathbf{u} \cdot \text{gyr}[\mathbf{x}, \mathbf{y}] \mathbf{v} = \mathbf{u} \cdot \mathbf{v}$ for all $\mathbf{u}, \mathbf{v} \in \mathbb{V}_c$. Einstein gyrovector space turns out to be a useful framework for hyperbolic geometry, since we recover the Klein model of curvature -1 for $\mathbb{V} = \mathbb{R}^n$ and $c = 1$.

4.2. Connecting the Klein Model with Einstein Gyrovector Spaces

Next, we show how Einstein gyrovector spaces facilitate the “tangent space construction” of scalar multiplication and vector addition in the Klein model. We start with the geodesics in the Klein model and their formulation using operations in the Einstein gyrovector spaces.

Geodesics The geodesic $\gamma_{\mathbf{x}^\mathcal{K} \rightarrow \mathbf{y}^\mathcal{K}} : \mathbb{R} \rightarrow \mathbb{K}^n$ connecting points $\mathbf{x}^\mathcal{K}, \mathbf{y}^\mathcal{K} \in \mathbb{K}^n$, such that $\gamma_{\mathbf{x}^\mathcal{K} \rightarrow \mathbf{y}^\mathcal{K}}(0) = \mathbf{x}^\mathcal{K}$ and $\gamma_{\mathbf{x}^\mathcal{K} \rightarrow \mathbf{y}^\mathcal{K}}(1) = \mathbf{y}^\mathcal{K}$, is shown by Ungar (2009, sec. 3.9) to be:

$$\gamma_{\mathbf{x}^\mathcal{K} \rightarrow \mathbf{y}^\mathcal{K}}(t) = \mathbf{x}^\mathcal{K} \oplus_E (\ominus_E \mathbf{x}^\mathcal{K} \oplus_E \mathbf{y}^\mathcal{K}) \otimes_E t, \quad (16)$$

where $\ominus_E \mathbf{x}^\mathcal{K}$ is the inverse of $\mathbf{x}^\mathcal{K}$ in the Einstein gyrogroup satisfying $\ominus_E \mathbf{x}^\mathcal{K} \oplus_E \mathbf{x}^\mathcal{K} = \mathbf{o}^\mathcal{K}$.

Scaling and Einstein scalar multiplication We study the tangent space construction of performing scalar multiplication in the Klein model and build the identity with the Einstein addition. The following theorem summarizes our finding:

Theorem 6 *For $\forall r \in \mathbb{R}, \mathbf{x}^\mathcal{K} \in \mathbb{K}^n$, performing scalar multiplication in the Klein model, which first uses the logarithmic map to project $\mathbf{x}^\mathcal{K}$ to $\mathcal{T}_{\mathbf{o}^\mathcal{K}} \mathbb{K}^n$, then multiplies this projection by a scalar in the tangent space and projects it back to the manifold with the exponential map, can be achieved by directly applying the Einstein scalar multiplication, namely,*

$$r \otimes_E \mathbf{x}^\mathcal{K} = \exp_{\mathbf{o}^\mathcal{K}}^\mathcal{K} (r \log_{\mathbf{o}^\mathcal{K}}^\mathcal{K} (\mathbf{x}^\mathcal{K})). \quad (17)$$

Parallel transport and Einstein addition We also connect the parallel transport with the Einstein addition using exponential and logarithmic maps in the following theorem:

Theorem 7 *The parallel transport w.r.t. the Levi-Civita connection of a tangent vector $\mathbf{v}^\mathcal{K} \in \mathcal{T}_{\mathbf{o}^\mathcal{K}} \mathbb{K}^n$ to another tangent space $\mathcal{T}_{\mathbf{x}^\mathcal{K}} \mathbb{K}^n$ in the Klein model is given by*

$$P_{\mathbf{o}^\mathcal{K} \rightarrow \mathbf{x}^\mathcal{K}}(\mathbf{v}^\mathcal{K}) = \log_{\mathbf{x}^\mathcal{K}}^\mathcal{K} (\mathbf{x}^\mathcal{K} \oplus_E \exp_{\mathbf{o}^\mathcal{K}}^\mathcal{K}(\mathbf{v}^\mathcal{K})). \quad (18)$$

4.3. Hyperbolic Neural Networks in the Klein Model

Theorems 6 and 7 provide compact formulas for defining neural operations for HNNs using the Klein model.

Weight matrix transformation and nonlinear activation First, thanks to Theorem 6, we define the “Einstein version” of Euclidean functions that preserve the origin.

Definition 8 (Einstein version) *Given $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ such that $f(\mathbf{0}) = (\mathbf{0})$, the Einstein version of f , denoted as $f^{\otimes_E} : \mathbb{K}^n \rightarrow \mathbb{K}^m$, is defined by*

$$f^{\otimes_E}(\mathbf{x}^\mathcal{K}) := \exp_{\mathbf{o}^\mathcal{K}}^\mathcal{K} (f(\log_{\mathbf{o}^\mathcal{K}}^\mathcal{K}(\mathbf{x}^\mathcal{K}))). \quad (19)$$

For instance, if $\sigma : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a non-linear activation function, then its Einstein version $\sigma^{\otimes_E}$ can be applied to points in the Klein model. Namely,

$$\sigma^{\otimes_E}(\mathbf{x}^{\mathcal{K}}) := \exp_{\mathbf{o}^{\mathcal{K}}}^{\mathcal{K}} \left(\sigma \left(\log_{\mathbf{o}^{\mathcal{K}}}^{\mathcal{K}}(\mathbf{x}^{\mathcal{K}}) \right) \right). \quad (20)$$

Moreover, we identify matrix-vector multiplication as an Einstein version of linear maps. Then the following result is straightforward.

Theorem 9 (Einstein matrix-vector multiplication) *If $\mathbf{M} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a linear map, which we identify with its matrix representation, then for $\mathbf{x}^{\mathcal{K}} \in \mathbb{K}^n$ s.t. $\mathbf{M}\mathbf{x}^{\mathcal{K}} \neq \mathbf{0}$,*

$$\begin{aligned} \mathbf{M}^{\otimes_E}(\mathbf{x}^{\mathcal{K}}) &= \exp_{\mathbf{o}^{\mathcal{K}}}^{\mathcal{K}}(\mathbf{M} \log_{\mathbf{o}^{\mathcal{K}}}^{\mathcal{K}}(\mathbf{x}^{\mathcal{K}})) \\ &= \tanh \left(\frac{2\|\mathbf{M}\mathbf{x}^{\mathcal{K}}\|}{\|\mathbf{x}^{\mathcal{K}}\|} \tanh^{-1} \left(\frac{\|\mathbf{x}^{\mathcal{K}}\|}{1 + \sqrt{1 - \|\mathbf{x}^{\mathcal{K}}\|^2}} \right) \right) \frac{\mathbf{M}\mathbf{x}^{\mathcal{K}}}{\|\mathbf{M}\mathbf{x}^{\mathcal{K}}\|}; \end{aligned} \quad (21)$$

and if $\mathbf{M}\mathbf{x}^{\mathcal{K}} = \mathbf{0}$, then $\mathbf{M}^{\otimes_E}(\mathbf{x}^{\mathcal{K}}) = \mathbf{0}$. Moreover, if we define the Einstein matrix-vector multiplication of $\mathbf{M} \in \mathcal{M}_{m,n}(\mathbb{R})$ and $\mathbf{x}^{\mathcal{K}} \in \mathbb{K}^n$ by $\mathbf{M} \otimes_E \mathbf{x}^{\mathcal{K}} := \mathbf{M}^{\otimes_E}(\mathbf{x}^{\mathcal{K}})$, then it shares common properties with the matrix-vector multiplication in the Euclidean space.

1. For $\mathbf{M} \in \mathcal{M}_{l,m}(\mathbb{R})$, $\mathbf{M}' \in \mathcal{M}_{m,n}(\mathbb{R})$, $(\mathbf{M}\mathbf{M}') \otimes_E \mathbf{x}^{\mathcal{K}} = \mathbf{M} \otimes_E (\mathbf{M}' \otimes_E \mathbf{x}^{\mathcal{K}})$;
2. For $r > 0$, $\mathbf{M} \in \mathcal{M}_{m,n}(\mathbb{R})$, $(r\mathbf{M}) \otimes_E \mathbf{x}^{\mathcal{K}} = r \otimes_E (\mathbf{M} \otimes_E \mathbf{x}^{\mathcal{K}})$;
3. For $\mathbf{M} \in \mathcal{O}_n(\mathbb{R})$, $\mathbf{M} \otimes_E \mathbf{x}^{\mathcal{K}} = \mathbf{M}\mathbf{x}^{\mathcal{K}}$.

Bias translation Next, Theorem 7 enables us to formalize bias translation in the Klein model. Specifically, we can use the Einstein addition in place of the bias translation of a point $\mathbf{x}^{\mathcal{K}} \in \mathbb{K}^n$ with a hyperbolic bias $\mathbf{b}^{\mathcal{K}} \in \mathbb{K}^n$,

$$\begin{aligned} \mathbf{x}^{\mathcal{K}} &\mapsto \exp_{\mathbf{x}^{\mathcal{K}}}^{\mathcal{K}} \left(P_{\mathbf{o}^{\mathcal{K}} \rightarrow \mathbf{x}^{\mathcal{K}}}^{\mathcal{K}} \left(\log_{\mathbf{o}^{\mathcal{K}}}^{\mathcal{K}}(\mathbf{b}^{\mathcal{K}}) \right) \right) \\ &= \exp_{\mathbf{x}^{\mathcal{K}}}^{\mathcal{K}} \left(\log_{\mathbf{x}^{\mathcal{K}}}^{\mathcal{K}} \left(\mathbf{x}^{\mathcal{K}} \oplus_E \exp_{\mathbf{o}^{\mathcal{K}}}^{\mathcal{K}} \left(\log_{\mathbf{o}^{\mathcal{K}}}^{\mathcal{K}}(\mathbf{b}^{\mathcal{K}}) \right) \right) \right) = \mathbf{x}^{\mathcal{K}} \oplus_E \mathbf{b}^{\mathcal{K}}. \end{aligned} \quad (22)$$

In summary, Einstein operations provide all we need to define fundamental neural operations in the Klein model.

5. Experiments

We perform node classification tasks on the well-known WebKB datasets including Texas, Wisconsin and Chameleon (Craven et al., 2000), Actor datasets (Tang et al., 2009) and citation datasets including Cora and Pubmed (Sen et al., 2008), summarized in Appendix C. These datasets enjoy high Gromov hyperbolicities (Jonckheere et al., 2008; Adcock et al., 2013). We compare HNNs using the three hyperbolic models we have discussed. We use HNNs containing a hyperbolic fullyconnected layer and a Euclidean classification layer. We use (hyperbolic version of) ReLU as the activation function. The HNNs are trained using the Riemannian Adam (Becigneul and Ganea, 2019) for 5000 epochs with early stopping. Our code is available at https://github.com/YidanM/Klein_HNN.

Table 1 presents the test accuracies. The results indicate that Klein HNNs performs on par with both HNNs using the Poincaré ball model and the hyperboloid model.

In Table 2, we report the average run time to train each epoch. Evidently, the average training time per epoch for the hyperboloid HNN is longer than those for Poincaré ball and

Table 1: The test accuracy of two-layer HNNs (mean and std over 3 trials)

Dataset	Texas	Wisconsin	Chameleon	Actor	Cora	Pubmed
Poincaré	0.9697±0.0347	0.9506±0.0283	0.7442±0.0064	0.6436±0.0057	0.5960±0.0101	0.7270±0.0036
Hyperboloid	0.9697±0.0132	0.9444±0.0321	0.7418±0.0120	0.6342±0.0226	0.6067±0.0137	0.7293±0.0064
Klein	0.9697±0.0263	0.9568±0.0214	0.7375±0.0119	0.6509±0.0067	0.5957±0.0106	0.7230±0.0139

Klein HNNs. This is because no simple formulation exists for the weight matrix transformation and bias translation in the hyperboloid model and we have to perform the tangent space operations which involve a series of exponential and logarithmic maps, which cause computational complexity. Thanks to the connection between the Klein model and Einstein gyrovector spaces, we are able to implement the Klein linear layer in a simple and elegant way. Empirically, Klein HNNs are as efficient as Poincaré HNNs and sometimes the most efficient among all the three models.

Table 2: Average runtime for training one epoch (mean and std over 3 trials)

Dataset	Texas	Wisconsin	Chameleon	Actor	Cora	Pubmed
Poincaré	0.0222±0.0025	0.0205±0.0008	0.0248±0.0008	0.0263±0.0009	0.0198±0.0010	0.0220±0.0023
Hyperboloid	0.0396±0.0087	0.0341±0.0012	0.0224±0.0021	0.0420±0.0014	0.0362±0.0022	0.0351±0.0014
Klein	0.0219±0.0017	0.0208±0.0020	0.0234±0.0029	0.0238±0.0008	0.0214±0.0012	0.0222±0.0017

We also plot the hyperbolic features, i.e., output of the hyperbolic linear layer. For illustration, we project them into the 2D-disk using hyperbolic t-SNE. Figure 1 shows the results on the Texas and Wisconsin datasets for Poincaré and Klein HNNs. Features with different labels are indicated by different colors. We observe similar classification results for the same dataset, indicating the effectiveness of our Klein HNNs.

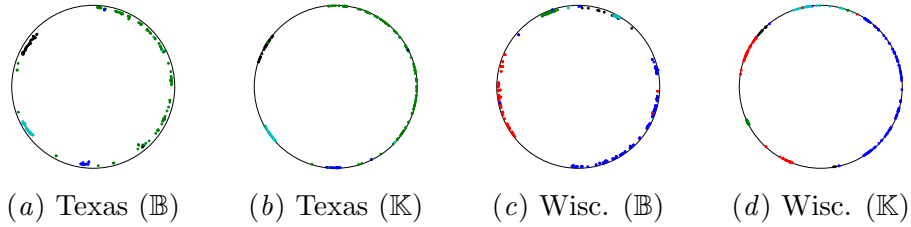


Figure 1: Features projected into the 2D-disks.

6. Conclusion

In this paper, we provide a detailed framework for Klein HNNs, addressing the gaps largely overlooked in previous works. We derive compact formulas for key operations in the Klein model. Subsequently, we connect the Riemannian geometry of the Klein model with the Einstein gyrovector spaces, providing a simple and elegant formulation for the tangent space construction of fully connected layers in the Klein model. Our experiments show that Klein HNNs achieve comparable performance with Poincaré/hyperboloid HNNs while maintaining training efficiency. Furthermore, our framework supports additional operations, such as Einstein midpoints, to build HNNs with more complicated architecture unified in the Klein model. Future works include extending the current operations to other common neural operations.

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Appendix A. Hyperbolic Geometry

We review basic facts about the Poincaré ball model and the hyperboloid model of hyperbolic space. For readers seeking a more introductory overview of hyperbolic geometry, we recommend [Anderson \(2006\)](#). We also recommend the work of [Peng et al. \(2021\)](#) for a survey of HNNs. For a comprehensive study of operations within these models, including those used in hyperbolic neural networks, we refer to the detailed work of [Ganea et al. \(2018a\)](#) and [Chami et al. \(2019\)](#).

A.1. The Poincaré Ball Model

The n -dimensional Poincaré ball model with constant negative curvature of -1 is the Riemannian manifold $\mathbb{B}^n = \{\mathbf{x} \in \mathbb{R}^n \mid \|\mathbf{x}\| < 1\}$ with the metric tensor $g_{\mathbf{x}}^{\mathbb{B}} = (\rho_{\mathbf{x}})^2 \mathbf{I}_n$, where $\rho_{\mathbf{x}} = 2(1 - \|\mathbf{x}\|^2)^{-1}$ is the conformal factor and $\mathbf{I}_n$ is the Euclidean metric. The geodesic distance between two point $\mathbf{x}, \mathbf{y} \in \mathbb{B}^n$ is given by

$$d_{\mathbb{B}}(\mathbf{x}, \mathbf{y}) = \cosh^{-1} \left(1 + \frac{2\|\mathbf{x} - \mathbf{y}\|^2}{(1 - \|\mathbf{x}\|^2)(1 - \|\mathbf{y}\|^2)} \right). \quad (23)$$

Given $\mathbf{x} \in \mathbb{B}^n$, $\mathcal{T}_{\mathbf{x}}\mathbb{B}^n$ denotes the tangent space of $\mathbb{B}^n$ at $\mathbf{x}$. For $\mathbf{x}, \mathbf{y} \in \mathbb{B}^n$ with $\mathbf{x} \neq \mathbf{y}$ and $\mathbf{v} \in \mathcal{T}_{\mathbf{x}}\mathbb{B}^n \setminus \{\mathbf{0}\}$, the exponential map $\exp_{\mathbf{x}}^{\mathbb{B}} : \mathcal{T}_{\mathbf{x}}\mathbb{B}^n \rightarrow \mathbb{B}^n$ and the logarithmic map $\log_{\mathbf{x}}^{\mathbb{B}} : \mathbb{B}^n \rightarrow \mathcal{T}_{\mathbf{x}}\mathbb{B}^n$ satisfy

$$\exp_{\mathbf{x}}^{\mathbb{B}}(\mathbf{v}) = \mathbf{x} \oplus_{\mathbf{M}} \left(\tanh \left(\frac{\rho_{\mathbf{x}} \|\mathbf{v}\|}{2} \right) \frac{\mathbf{v}}{\|\mathbf{v}\|} \right), \quad (24)$$

$$\log_{\mathbf{x}}^{\mathbb{B}}(\mathbf{y}) = \frac{2}{\rho_{\mathbf{x}}} \tanh^{-1}(\|\mathbf{x} \oplus_{\mathbf{M}} \mathbf{y}\|) \frac{-\mathbf{x} \oplus_{\mathbf{M}} \mathbf{y}}{\|\mathbf{x} \oplus_{\mathbf{M}} \mathbf{y}\|}, \quad (25)$$

where $\oplus_{\mathbf{M}}$ is the Möbius addition, defined by

$$\mathbf{x} \oplus_{\mathbf{M}} \mathbf{y} = \frac{(1 + 2\mathbf{x} \cdot \mathbf{y} + \|\mathbf{y}\|^2) \mathbf{x} + (1 - \|\mathbf{x}\|^2) \mathbf{y}}{1 + 2\mathbf{x} \cdot \mathbf{y} + \|\mathbf{x}\|^2 \|\mathbf{y}\|^2}. \quad (26)$$

In particular, the exponential and logarithmic maps at the hyperbolic origin $\mathbf{o}$ are given by

$$\exp_{\mathbf{o}}^{\mathbb{B}}(\mathbf{v}) = \tanh(\|\mathbf{v}\|) \frac{\mathbf{v}}{\|\mathbf{v}\|}, \quad \log_{\mathbf{o}}^{\mathbb{B}}(\mathbf{y}) = \tanh^{-1}(\|\mathbf{y}\|) \frac{\mathbf{y}}{\|\mathbf{y}\|}. \quad (27)$$

[Ganea et al. \(2018a\)](#) proved that, the parallel transport w.r.t. the Levi-Civita connection of $\mathbf{v} \in \mathcal{T}_{\mathbf{o}}\mathbb{B}^n$ to the tangent space $\mathcal{T}_{\mathbf{x}}\mathbb{B}^n$ at $\mathbf{x}$ is given by

$$P_{\mathbf{o} \rightarrow \mathbf{x}}(\mathbf{v}) = \log_{\mathbf{x}}^{\mathbb{B}}(\mathbf{x} \oplus_{\mathbf{M}} \exp_{\mathbf{o}}^{\mathbb{B}}(\mathbf{v})). \quad (28)$$

A.2. The Hyperboloid Model

The n -dimensional hyperboloid model with constant negative curvature -1 is represented as $\mathbb{L}^n = \{\mathbf{x} \in \mathbb{R}^{n+1} \mid \langle \mathbf{x}, \mathbf{x} \rangle_{\mathcal{L}} = -1, x_t > 0\}$, where $\langle \cdot, \cdot \rangle_{\mathcal{L}}$ denotes the Minkowski inner product, $\langle \mathbf{x}, \mathbf{y} \rangle_{\mathcal{L}} := -x_0 y_0 + x_1 y_1 + \dots + x_n y_n$. It has the metric tensor $g^{\mathbb{L}} = \text{diag}([-1, \mathbf{1}_n^{\top}])$. From a special relativity perspective, we may also write $\mathbf{x} \in \mathbb{L}^n$ as $[x_t, \mathbf{x}_s^{\top}]^{\top}$, for x_t being the time axis and $\mathbf{x}_s$ being the spatial axes.

The tangent space of $\mathbb{L}^n$ at $\mathbf{x}$ is the orthogonal space of $\mathbb{L}^n$ with respect to the Minkowski inner product and is represented as $\mathcal{T}_{\mathbf{x}}\mathbb{L}^n := \{\mathbf{v} \in \mathbb{R}^{n+1} : \langle \mathbf{v}, \mathbf{x} \rangle_{\mathcal{L}} = 0\}$. Specifically, let $\{\mathbf{e}_0, \dots, \mathbf{e}_n\}$ be the canonical basis of $\mathbb{R}^{n+1}$, the chart-induced basis for the tangent space at $\mathbf{x}$ is can be written as

$$\left. \frac{\partial}{\partial x_i} \right|_{\mathbf{x}} = \frac{\mathbf{x} \cdot \mathbf{e}_i}{\sqrt{1 + \|\mathbf{x}\|^2}} \mathbf{e}_0 + \mathbf{e}_i. \quad (29)$$

The unit-speed geodesic $\phi_{\mathbf{x}, \mathbf{v}}(t)$ with $\phi_{\mathbf{x}, \mathbf{v}}(0) = \mathbf{x} \in \mathbb{L}^n$ and $\dot{\phi}_{\mathbf{x}, \mathbf{v}}(0) = \mathbf{v} \in \mathcal{T}_{\mathbf{x}}\mathbb{L}^n$ is given by

$$\phi_{\mathbf{x}, \mathbf{v}}(t) = \mathbf{x} \cosh(t) + \mathbf{v} \sinh(t). \quad (30)$$

The geodesic distance of $\mathbf{x}, \mathbf{y} \in \mathbb{L}^n$ is given by $d_{\mathcal{L}}(\mathbf{x}, \mathbf{y}) = \cosh^{-1}(-\langle \mathbf{x}, \mathbf{y} \rangle_{\mathcal{L}})$. For $\mathbf{x}, \mathbf{y} \in \mathbb{L}^n$ with $\mathbf{x} \neq \mathbf{y}$ and $\mathbf{v} \in \mathcal{T}_{\mathbf{x}}\mathbb{L}^n \setminus \{\mathbf{0}\}$, the exponential map $\exp_{\mathbf{x}}^{\mathcal{L}} : \mathcal{T}_{\mathbf{x}}\mathbb{L}^n \rightarrow \mathbb{L}^n$ and the logarithmic map $\log_{\mathbf{x}}^{\mathcal{L}} : \mathbb{L}^n \rightarrow \mathcal{T}_{\mathbf{x}}\mathbb{L}^n$ are given by

$$\exp_{\mathbf{x}}^{\mathcal{L}}(\mathbf{v}) = \cosh(\|\mathbf{v}\|_{\mathcal{L}}) \mathbf{x} + \sinh(\|\mathbf{v}\|_{\mathcal{L}}) \frac{\mathbf{v}}{\|\mathbf{v}\|_{\mathcal{L}}}, \quad (31)$$

$$\log_{\mathbf{x}}^{\mathcal{L}}(\mathbf{y}) = d_{\mathcal{L}}(\mathbf{x}, \mathbf{y}) \frac{\mathbf{y} + \langle \mathbf{x}, \mathbf{y} \rangle_{\mathcal{L}} \mathbf{x}}{\|\mathbf{y} + \langle \mathbf{x}, \mathbf{y} \rangle_{\mathcal{L}} \mathbf{x}\|_{\mathcal{L}}}, \quad (32)$$

where $\|\mathbf{v}\|_{\mathcal{L}} = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle_{\mathcal{L}}}$ is the induced norm from the Minkowski inner product.

The parallel transport of a tangent vector $\mathbf{v} \in \mathcal{T}_{\mathbf{o}}\mathbb{L}^n$ at the hyperbolic origin $\mathbf{o} = [1, \mathbf{0}^{\top}]^{\top}$ to the tangent space $\mathcal{T}_{\mathbf{x}}\mathbb{L}^n$ is given by

$$P_{\mathbf{o} \rightarrow \mathbf{x}}(\mathbf{v}) = \mathbf{v} - \frac{\langle \log_{\mathbf{o}}(\mathbf{x}), \mathbf{v} \rangle_{\mathcal{L}}}{d_{\mathcal{L}}(\mathbf{o}, \mathbf{x})^2} (\log_{\mathbf{o}}(\mathbf{x}) + \log_{\mathbf{x}}(\mathbf{o})). \quad (33)$$

Appendix B. Additional Proofs

B.1. Proof of Lemma 1

1. The corresponding tangent vectors $\mathbf{v}^{\mathcal{B}} \in \mathcal{T}_{\mathbf{x}^{\mathcal{B}}}\mathbb{B}^n$ and $\mathbf{v}^{\mathcal{K}} \in \mathcal{T}_{\mathbf{x}^{\mathcal{K}}}\mathbb{K}^n$ are related using Jacobian matrices. Namely,

$$\mathbf{v}^{\mathcal{K}} = \frac{\partial \pi_{\mathcal{B} \rightarrow \mathcal{K}}(\mathbf{x}^{\mathcal{B}})}{\partial \mathbf{x}^{\mathcal{B}}} \mathbf{v}^{\mathcal{B}}, \quad \mathbf{v}^{\mathcal{B}} = \frac{\partial \pi_{\mathcal{K} \rightarrow \mathcal{B}}(\mathbf{x}^{\mathcal{K}})}{\partial \mathbf{x}^{\mathcal{K}}} \mathbf{v}^{\mathcal{K}}. \quad (34)$$

Since

$$\frac{\partial \pi_{\mathcal{B} \rightarrow \mathcal{K}}(\mathbf{x}^{\mathcal{B}})}{\partial \mathbf{x}^{\mathcal{B}}} = \frac{2}{1 + \|\mathbf{x}^{\mathcal{B}}\|^2} \mathbf{I} + \mathbf{x}^{\mathcal{B}} \cdot \frac{-2 \cdot 2(\mathbf{x}^{\mathcal{B}})^{\top}}{(1 + \|\mathbf{x}^{\mathcal{B}}\|^2)^2}, \quad (35)$$

$$\frac{\partial \pi_{\mathcal{K} \rightarrow \mathcal{B}}(\mathbf{x}^{\mathcal{K}})}{\partial \mathbf{x}^{\mathcal{K}}} = \frac{1}{1 + \sqrt{1 - \|\mathbf{x}^{\mathcal{K}}\|^2}} \mathbf{I} + \mathbf{x}^{\mathcal{K}} \cdot \frac{-1 \cdot 2(\mathbf{x}^{\mathcal{K}})^{\top}}{2\sqrt{1 - \|\mathbf{x}^{\mathcal{K}}\|^2}(1 + \sqrt{1 - \|\mathbf{x}^{\mathcal{K}}\|^2})^2}, \quad (36)$$

we have

$$\mathbf{v}^{\mathcal{K}} = \frac{\partial \pi_{\mathcal{B} \rightarrow \mathcal{K}}(\mathbf{x}^{\mathcal{B}})}{\partial \mathbf{x}^{\mathcal{B}}} \mathbf{v}^{\mathcal{B}} = \frac{2}{1 + \|\mathbf{x}^{\mathcal{B}}\|^2} \mathbf{v}^{\mathcal{B}} - \frac{4\mathbf{x}^{\mathcal{B}} \cdot \mathbf{v}^{\mathcal{B}}}{(1 + \|\mathbf{x}^{\mathcal{B}}\|^2)^2} \mathbf{x}^{\mathcal{B}}, \quad (37)$$

$$\mathbf{v}^{\mathcal{K}} = \frac{\partial \pi_{\mathcal{K} \rightarrow \mathcal{B}}(\mathbf{x}^{\mathcal{K}})}{\partial \mathbf{x}^{\mathcal{K}}} \mathbf{v}^{\mathcal{K}} = \frac{1}{1 + \sqrt{1 - \|\mathbf{x}^{\mathcal{K}}\|^2}} \mathbf{v}^{\mathcal{K}} + \frac{\mathbf{x}^{\mathcal{K}} \cdot \mathbf{v}^{\mathcal{K}}}{\sqrt{1 - \|\mathbf{x}^{\mathcal{K}}\|^2}(1 + \sqrt{1 - \|\mathbf{x}^{\mathcal{K}}\|^2})^2} \mathbf{x}^{\mathcal{K}}. \quad (38)$$

2. Similarly,

$$\mathbf{v}^{\mathcal{K}} = \frac{\partial \pi_{\mathcal{L} \rightarrow \mathcal{K}}(\mathbf{x}^{\mathcal{L}})}{\partial \mathbf{x}^{\mathcal{L}}} \mathbf{v}^{\mathcal{L}}, \quad \mathbf{v}^{\mathcal{L}} = \frac{\partial \pi_{\mathcal{K} \rightarrow \mathcal{L}}(\mathbf{x}^{\mathcal{K}})}{\partial \mathbf{x}^{\mathcal{K}}} \mathbf{v}^{\mathcal{K}}. \quad (39)$$

For $\frac{\partial \pi_{\mathcal{L} \rightarrow \mathcal{K}}(\mathbf{x}^{\mathcal{L}})}{\partial \mathbf{x}^{\mathcal{L}}}$, let $\mathbf{y} = \pi_{\mathcal{L} \rightarrow \mathcal{K}}(\mathbf{x}^{\mathcal{L}})$. Then

$$\frac{\partial \pi_{\mathcal{L} \rightarrow \mathcal{K}}(\mathbf{x}^{\mathcal{L}})}{\partial \mathbf{x}^{\mathcal{L}}} = \frac{\partial \mathbf{y}}{\partial \mathbf{x}^{\mathcal{L}}} = \begin{bmatrix} \frac{\partial \mathbf{y}}{\partial x_t} & \frac{\partial \mathbf{y}}{\partial \mathbf{x}_s} \end{bmatrix}, \quad (40)$$

with $\frac{\partial \mathbf{y}}{\partial x_t}$ being a $n \times 1$ column vector and $\frac{\partial \mathbf{y}}{\partial \mathbf{x}_s}$ being a $n \times n$ matrix. The isometric mapping in Equation (2) indicates that $\mathbf{y} = \frac{1}{x_t} \mathbf{x}_s$, and thus

$$\frac{\partial \mathbf{y}}{\partial x_t} = -\frac{1}{x_t^2} \mathbf{x}_s, \quad \frac{\partial \mathbf{y}}{\partial \mathbf{x}_s} = \frac{1}{x_t} \mathbf{I}. \quad (41)$$

Writing $\mathbf{v}^{\mathcal{L}}$ as $[v_t, \mathbf{v}_s^\top]^\top$, we have

$$\mathbf{v}^{\mathcal{K}} = \frac{\partial \pi_{\mathcal{L} \rightarrow \mathcal{K}}(\mathbf{x}^{\mathcal{L}})}{\partial \mathbf{x}^{\mathcal{L}}} \mathbf{v}^{\mathcal{L}} = \begin{bmatrix} \frac{\partial \mathbf{y}}{\partial x_t} & \frac{\partial \mathbf{y}}{\partial \mathbf{x}_s} \end{bmatrix} \begin{bmatrix} v_t \\ \mathbf{v}_s \end{bmatrix} = v_t \frac{\partial \mathbf{y}}{\partial x_t} + \frac{\partial \mathbf{y}}{\partial \mathbf{x}_s} \mathbf{v}_s = -\frac{v_t}{x_t^2} \mathbf{x}_s + \frac{1}{x_t} \mathbf{v}_s. \quad (42)$$

Hence,

$$\mathbf{v}^{\mathcal{K}} = \frac{\partial \pi_{\mathcal{L} \rightarrow \mathcal{K}}(\mathbf{x}^{\mathcal{L}})}{\partial \mathbf{x}^{\mathcal{L}}} \mathbf{v}^{\mathcal{L}} = -\frac{v_t}{x_t^2} \mathbf{x}_s + \frac{1}{x_t} \mathbf{v}_s \in \mathcal{T}_{\mathbf{x}^{\mathcal{K}}} \mathbb{K}^n. \quad (43)$$

For $\frac{\partial \pi_{\mathcal{K} \rightarrow \mathcal{L}}(\mathbf{x}^{\mathcal{K}})}{\partial \mathbf{x}^{\mathcal{K}}}$, let $\mathbf{z} = \pi_{\mathcal{K} \rightarrow \mathcal{L}}(\mathbf{x}^{\mathcal{K}}) = [z_t, \mathbf{z}_s^\top]^\top$. Then

$$\frac{\partial \pi_{\mathcal{K} \rightarrow \mathcal{L}}(\mathbf{x}^{\mathcal{K}})}{\partial \mathbf{x}^{\mathcal{K}}} = \frac{\partial \mathbf{z}}{\partial \mathbf{x}^{\mathcal{K}}} = \begin{bmatrix} \frac{\partial z_t}{\partial \mathbf{x}^{\mathcal{K}}} \\ \frac{\partial \mathbf{z}_s}{\partial \mathbf{x}^{\mathcal{K}}} \end{bmatrix}. \quad (44)$$

with $\frac{\partial z_t}{\partial \mathbf{x}^{\mathcal{K}}}$ being a $1 \times n$ row vector and $\frac{\partial \mathbf{z}_s}{\partial \mathbf{x}^{\mathcal{K}}}$ being a $n \times n$ matrix. From the mapping between the hyperboloid model and the Klein model, we know

$$z_t = \frac{1}{\sqrt{1 - \|\mathbf{x}^{\mathcal{K}}\|^2}}, \quad \mathbf{z}_s = \frac{1}{\sqrt{1 - \|\mathbf{x}^{\mathcal{K}}\|^2}} \mathbf{x}^{\mathcal{K}}. \quad (45)$$

We compute $\frac{\partial z_t}{\partial \mathbf{x}^{\mathcal{K}}}$ and $\frac{\partial \mathbf{z}_s}{\partial \mathbf{x}^{\mathcal{K}}}$ respectively:

$$\frac{\partial z_t}{\partial \mathbf{x}^{\mathcal{K}}} = -\frac{1}{2} (1 - \|\mathbf{x}^{\mathcal{K}}\|^2)^{-\frac{3}{2}} \cdot (-2(\mathbf{x}^{\mathcal{K}})^\top) = \frac{(\mathbf{x}^{\mathcal{K}})^\top}{(1 - \|\mathbf{x}^{\mathcal{K}}\|^2)^{\frac{3}{2}}}; \quad (46)$$

$$\frac{\partial \mathbf{z}_s}{\partial \mathbf{x}^{\mathcal{K}}} = \frac{1}{\sqrt{1 - \|\mathbf{x}^{\mathcal{K}}\|^2}} \mathbf{I} + \mathbf{x}^{\mathcal{K}} \cdot \frac{(\mathbf{x}^{\mathcal{K}})^\top}{(1 - \|\mathbf{x}^{\mathcal{K}}\|^2)^{\frac{3}{2}}}. \quad (47)$$

Putting everything together,

$$\mathbf{v}^{\mathcal{L}} = \frac{\partial \pi_{\mathcal{K} \rightarrow \mathcal{L}}(\mathbf{x}^{\mathcal{K}})}{\partial \mathbf{x}^{\mathcal{K}}} \mathbf{v}^{\mathcal{K}} = \begin{bmatrix} \frac{\mathbf{x}^{\mathcal{K}} \cdot \mathbf{v}^{\mathcal{K}}}{(1 - \|\mathbf{x}^{\mathcal{K}}\|^2)^{\frac{3}{2}}} \\ \frac{1}{\sqrt{1 - \|\mathbf{x}^{\mathcal{K}}\|^2}} \mathbf{v}^{\mathcal{K}} + \frac{\mathbf{x}^{\mathcal{K}} \cdot \mathbf{v}^{\mathcal{K}}}{(1 - \|\mathbf{x}^{\mathcal{K}}\|^2)^{\frac{3}{2}}} \mathbf{x}^{\mathcal{K}} \end{bmatrix} \in \mathcal{T}_{\mathbf{x}^{\mathcal{L}}} \mathbb{L}^n. \quad (48)$$

■

B.2. Proof of Lemma 3

The proof is analogous to that of $\mathbb{B}^n$ (Ganea et al., 2018b, Theorem 1). Let $\mathbf{x}^\mathcal{L} \in \mathbb{L}^n$ and $\mathbf{v}^\mathcal{L} \in \mathcal{T}_{\mathbf{x}^\mathcal{L}}\mathbb{L}^n$ with $\langle \mathbf{v}^\mathcal{L}, \mathbf{v}^\mathcal{L} \rangle_\mathcal{L} = 1$. According to Equation (30), the unit-speed geodesic $\phi(t) = \phi_{\mathbf{x}^\mathcal{L}, \mathbf{v}^\mathcal{L}}(t)$ with $\phi_{\mathbf{x}^\mathcal{L}, \mathbf{v}^\mathcal{L}}(0) = \mathbf{x}^\mathcal{L}$ and $\dot{\phi}_{\mathbf{x}^\mathcal{L}, \mathbf{v}^\mathcal{L}}(0) = \mathbf{v}^\mathcal{L}$ is

$$\phi_{\mathbf{x}^\mathcal{L}, \mathbf{v}^\mathcal{L}}(t) = \mathbf{x}^\mathcal{L} \cosh(t) + \mathbf{v}^\mathcal{L} \sinh(t). \quad (49)$$

Let $\mathbf{x}^\mathcal{K} \in \mathbb{K}^n$ and $\mathbf{v}^\mathcal{K} \in \mathcal{T}_{\mathbf{x}^\mathcal{K}}\mathbb{K}^n$ with $g_{\mathbf{x}^\mathcal{K}}(\mathbf{v}^\mathcal{K}, \mathbf{v}^\mathcal{K}) = 1$. Let $\gamma(t) = \gamma_{\mathbf{x}^\mathcal{K}, \mathbf{v}^\mathcal{K}}(t) \in \mathbb{K}^n$ denote the unit-speed geodesic in $\mathbb{K}^n$ with $\gamma_{\mathbf{x}^\mathcal{K}, \mathbf{v}^\mathcal{K}}(0) = \mathbf{x}^\mathcal{K}$ and $\dot{\gamma}_{\mathbf{x}^\mathcal{K}, \mathbf{v}^\mathcal{K}}(0) = \mathbf{v}^\mathcal{K}$. Suppose $\mathbf{x}^\mathcal{L}$ and $\mathbf{x}^\mathcal{K}$ are corresponding points in $\mathbb{L}^n$ and $\mathbf{v}^\mathcal{L}$ and $\mathbf{v}^\mathcal{K}$ are corresponding vectors in their tangent spaces. Then

$$\mathbf{x}^\mathcal{L} = \pi_{\mathcal{K} \rightarrow \mathcal{L}}(\mathbf{x}^\mathcal{K}) = \frac{1}{\sqrt{1 - \|\mathbf{x}^\mathcal{K}\|^2}} \begin{bmatrix} 1 \\ \mathbf{x}^\mathcal{K} \end{bmatrix} = \lambda_{\mathbf{x}^\mathcal{K}} \begin{bmatrix} 1 \\ \mathbf{x}^\mathcal{K} \end{bmatrix}, \quad (50)$$

$$\mathbf{v}^\mathcal{L} = \dot{\phi}(0) = \left. \frac{\partial \pi_{\mathcal{K} \rightarrow \mathcal{L}}(\mathbf{y}^\mathcal{K})}{\partial \mathbf{y}^\mathcal{K}} \right|_{\gamma(0)} \dot{\gamma}(0) = \frac{\partial \pi_{\mathcal{K} \rightarrow \mathcal{L}}(\mathbf{x}^\mathcal{K})}{\partial \mathbf{x}^\mathcal{K}} \mathbf{v}^\mathcal{K}. \quad (51)$$

With

$$\frac{\partial \lambda_{\mathbf{x}^\mathcal{K}}}{\partial \mathbf{x}^\mathcal{K}} = \frac{\mathbf{x}^\mathcal{K}}{(1 - \|\mathbf{x}^\mathcal{K}\|^2)^{3/2}} = \lambda_{\mathbf{x}^\mathcal{K}}^3 \mathbf{x}^\mathcal{K}, \quad (52)$$

it follows that

$$\begin{aligned} \mathbf{v}^\mathcal{L} &= \frac{\partial \pi_{\mathcal{K} \rightarrow \mathcal{L}}(\mathbf{x}^\mathcal{K})}{\partial \mathbf{x}^\mathcal{K}} \mathbf{v}^\mathcal{K} = \left(\frac{\partial}{\partial \mathbf{x}^\mathcal{K}} \left(\begin{bmatrix} \lambda_{\mathbf{x}^\mathcal{K}} \\ \lambda_{\mathbf{x}^\mathcal{K}} \mathbf{x}^\mathcal{K} \end{bmatrix} \right) \right) \mathbf{v}^\mathcal{K} = \begin{bmatrix} \frac{\partial \lambda_{\mathbf{x}^\mathcal{K}}}{\partial \mathbf{x}^\mathcal{K}} \\ \frac{\partial \lambda_{\mathbf{x}^\mathcal{K}}}{\partial \mathbf{x}^\mathcal{K}} (\mathbf{x}^\mathcal{K})^\top + \lambda_{\mathbf{x}^\mathcal{K}} \mathbf{I} \end{bmatrix} \mathbf{v}^\mathcal{K} \\ &= \begin{bmatrix} \lambda_{\mathbf{x}^\mathcal{K}}^3 (\mathbf{x}^\mathcal{K} \cdot \mathbf{v}^\mathcal{K}) \\ \lambda_{\mathbf{x}^\mathcal{K}}^3 (\mathbf{x}^\mathcal{K} \cdot \mathbf{v}^\mathcal{K}) \mathbf{x}^\mathcal{K} + \lambda_{\mathbf{x}^\mathcal{K}} \mathbf{v}^\mathcal{K} \end{bmatrix}. \end{aligned} \quad (53)$$

Making substitutions in Equation (49) yields

$$\phi(t) = \lambda_{\mathbf{x}^\mathcal{K}} \begin{bmatrix} 1 \\ \mathbf{x}^\mathcal{K} \end{bmatrix} \cosh(t) + \begin{bmatrix} \lambda_{\mathbf{x}^\mathcal{K}}^3 (\mathbf{x}^\mathcal{K} \cdot \mathbf{v}^\mathcal{K}) \\ \lambda_{\mathbf{x}^\mathcal{K}}^3 (\mathbf{x}^\mathcal{K} \cdot \mathbf{v}^\mathcal{K}) \mathbf{x}^\mathcal{K} + \lambda_{\mathbf{x}^\mathcal{K}} \mathbf{v}^\mathcal{K} \end{bmatrix} \sinh(t). \quad (54)$$

Since the geodesics are preserved by the isometric mapping, γ is given by

$$\begin{aligned} \gamma_{\mathbf{x}^\mathcal{K}, \mathbf{v}^\mathcal{K}}(t) &= \pi_{\mathcal{L} \rightarrow \mathcal{K}} \circ \phi(t) \\ &= \frac{(\lambda_{\mathbf{x}^\mathcal{K}} \cosh(t) + \lambda_{\mathbf{x}^\mathcal{K}}^3 (\mathbf{x}^\mathcal{K} \cdot \mathbf{v}^\mathcal{K}) \sinh(t)) \mathbf{x}^\mathcal{K} + (\lambda_{\mathbf{x}^\mathcal{K}} \sinh(t)) \mathbf{v}^\mathcal{K}}{\lambda_{\mathbf{x}^\mathcal{K}} \cosh(t) + \lambda_{\mathbf{x}^\mathcal{K}}^3 (\mathbf{x}^\mathcal{K} \cdot \mathbf{v}^\mathcal{K}) \sinh(t)} \\ &= \frac{(\cosh(t) + \lambda_{\mathbf{x}^\mathcal{K}}^2 (\mathbf{x}^\mathcal{K} \cdot \mathbf{v}^\mathcal{K}) \sinh(t)) \mathbf{x}^\mathcal{K} + \sinh(t) \mathbf{v}^\mathcal{K}}{\cosh(t) + \lambda_{\mathbf{x}^\mathcal{K}}^2 (\mathbf{x}^\mathcal{K} \cdot \mathbf{v}^\mathcal{K}) \sinh(t)} \\ &= \mathbf{x}^\mathcal{K} + \frac{\sinh(t) \mathbf{v}^\mathcal{K}}{\cosh(t) + \lambda_{\mathbf{x}^\mathcal{K}}^2 (\mathbf{x}^\mathcal{K} \cdot \mathbf{v}^\mathcal{K}) \sinh(t)}. \end{aligned} \quad (55)$$

■

B.3. Proof of Proposition 5

Write $\mathbf{x}^\mathcal{L}$ as $[x_t, \mathbf{x}_s^\top]^\top$. Since $\mathbf{o}^\mathcal{L} = [1, 0, \dots, 0]^\top$, we have $d_\mathcal{L}(\mathbf{o}^\mathcal{L}, \mathbf{x}^\mathcal{L}) = \cosh^{-1}(x_t)$. Then

$$\log_{\mathbf{o}^\mathcal{L}}(\mathbf{x}^\mathcal{L}) = d_\mathcal{L}(\mathbf{o}^\mathcal{L}, \mathbf{x}^\mathcal{L}) \frac{\mathbf{x}^\mathcal{L} + \langle \mathbf{o}^\mathcal{L}, \mathbf{x}^\mathcal{L} \rangle_\mathcal{L} \mathbf{o}^\mathcal{L}}{\|\mathbf{x}^\mathcal{L} + \langle \mathbf{o}^\mathcal{L}, \mathbf{x}^\mathcal{L} \rangle_\mathcal{L} \mathbf{o}^\mathcal{L}\|_\mathcal{L}} = \cosh^{-1}(x_t) \frac{\mathbf{x}^\mathcal{L} - x_t \mathbf{o}^\mathcal{L}}{\|\mathbf{x}^\mathcal{L} - x_t \mathbf{o}^\mathcal{L}\|_\mathcal{L}}. \quad (56)$$

Here, $\mathbf{x}^\mathcal{L} - x_t \mathbf{o}^\mathcal{L} = [x_t, \mathbf{x}_s^\top]^\top - [x_t, \mathbf{0}^\top]^\top = [0, \mathbf{x}_s^\top]^\top$ and thus $\|\mathbf{x}^\mathcal{L} - x_t \mathbf{o}^\mathcal{L}\|_\mathcal{L} = \|\mathbf{x}_s\|$. Therefore, $\log_{\mathbf{o}^\mathcal{L}}(\mathbf{x}^\mathcal{L}) = \frac{\cosh^{-1}(x_t)}{\|\mathbf{x}_s\|} [0, \mathbf{x}_s^\top]^\top$. Similarly,

$$\log_{\mathbf{x}^\mathcal{L}}(\mathbf{o}^\mathcal{L}) = d_\mathcal{L}(\mathbf{x}^\mathcal{L}, \mathbf{o}^\mathcal{L}) \frac{\mathbf{o}^\mathcal{L} + \langle \mathbf{x}^\mathcal{L}, \mathbf{o}^\mathcal{L} \rangle_\mathcal{L} \mathbf{x}^\mathcal{L}}{\|\mathbf{o}^\mathcal{L} + \langle \mathbf{x}^\mathcal{L}, \mathbf{o}^\mathcal{L} \rangle_\mathcal{L} \mathbf{x}^\mathcal{L}\|_\mathcal{L}} = \cosh^{-1}(x_t) \frac{\mathbf{o}^\mathcal{L} - x_t \mathbf{x}^\mathcal{L}}{\|\mathbf{o}^\mathcal{L} - x_t \mathbf{x}^\mathcal{L}\|_\mathcal{L}}. \quad (57)$$

Here, $\mathbf{o}^\mathcal{L} - x_t \mathbf{x}^\mathcal{L} = [1, \mathbf{0}^\top]^\top - x_t [x_t, \mathbf{x}_s^\top]^\top = [1 - x_t^2, x_t \mathbf{x}_s^\top]^\top$, and thus

$$\begin{aligned} \|\mathbf{o}^\mathcal{L} - x_t \mathbf{x}^\mathcal{L}\|_\mathcal{L} &= \sqrt{-(1 - x_t^2)^2 + x_t^2 \|\mathbf{x}_s\|^2} = \sqrt{-(-\|\mathbf{x}_s\|^2)^2 + x_t^2 \|\mathbf{x}_s\|^2} \\ &= \sqrt{\|\mathbf{x}_s\|^2 (x_t^2 - \|\mathbf{x}_s\|^2)} = \sqrt{\|\mathbf{x}_s\|^2} = \|\mathbf{x}_s\|. \end{aligned} \quad (58)$$

Therefore, $\log_{\mathbf{x}^\mathcal{L}}(\mathbf{o}^\mathcal{L}) = \frac{\cosh^{-1}(x_t)}{\|\mathbf{x}_s\|} [1 - x_t^2, x_t \mathbf{x}_s^\top]^\top$. Since $\mathbf{v}^\mathcal{L} \in \mathcal{T}_{\mathbf{o}^\mathcal{L}} \mathbb{L}^n$, it holds that $\langle \mathbf{o}, \mathbf{v}^\mathcal{L} \rangle_\mathcal{L} = 0$. Writing $\mathbf{v}^\mathcal{L}$ as $[v_t, \mathbf{v}_s^\top]^\top$, this indicates that $v_t = 0$. Hence,

$$\begin{aligned} P_{\mathbf{o}^\mathcal{L} \rightarrow \mathbf{x}^\mathcal{L}}^\mathcal{L}(\mathbf{v}^\mathcal{L}) &= \mathbf{v}^\mathcal{L} - \frac{\langle \log_{\mathbf{o}^\mathcal{L}}(\mathbf{x}^\mathcal{L}), \mathbf{v}^\mathcal{L} \rangle_\mathcal{L}}{d_\mathcal{L}(\mathbf{o}^\mathcal{L}, \mathbf{x}^\mathcal{L})^2} (\log_{\mathbf{o}^\mathcal{L}}^\mathcal{L}(\mathbf{x}^\mathcal{L}) + \log_{\mathbf{x}^\mathcal{L}}^\mathcal{L}(\mathbf{o}^\mathcal{L})) \\ &= \begin{bmatrix} 0 \\ \mathbf{v}_s \end{bmatrix} - \frac{\frac{\cosh^{-1}(x_t)}{\|\mathbf{x}_s\|} \mathbf{x}_s \cdot \mathbf{v}_s \cosh^{-1}(x_t)}{(\cosh^{-1}(x_t))^2 \|\mathbf{x}_s\|} \left(\begin{bmatrix} 0 \\ \mathbf{x}_s \end{bmatrix} + \begin{bmatrix} 1 - x_t^2 \\ x_t \mathbf{x}_s \end{bmatrix} \right) \\ &= \begin{bmatrix} 0 \\ \mathbf{v}_s \end{bmatrix} - \frac{\mathbf{x}_s \cdot \mathbf{v}_s}{\|\mathbf{x}_s\|^2} \begin{bmatrix} 1 - x_t^2 \\ (1 + x_t) \mathbf{x}_s \end{bmatrix} = \begin{bmatrix} \mathbf{x}_s \cdot \mathbf{v}_s \\ \mathbf{v}_s - \frac{\mathbf{x}_s \cdot \mathbf{v}_s}{(x_t - 1)} \mathbf{x}_s \end{bmatrix}. \end{aligned} \quad (59)$$

Next, we use the isometric mapping in Equation (2) as well as Equations (6) and (7) to derive $P_{\mathbf{o}^\mathcal{K} \rightarrow \mathbf{x}^\mathcal{K}}(\mathbf{v}^\mathcal{K})$. For clarity, denote $P_{\mathbf{o}^\mathcal{L} \rightarrow \mathbf{x}^\mathcal{L}}(\mathbf{v}^\mathcal{L}) = \mathbf{w}^\mathcal{L} = [w_t, \mathbf{w}_s^\top]^\top \in \mathcal{T}_{\mathbf{x}^\mathcal{L}} \mathbb{L}^n$. Then, by Lemma 1, the corresponding tangent vector in $\mathcal{T}_{\mathbf{x}^\mathcal{K}} \mathbb{K}^n$ is given by

$$\mathbf{w}^\mathcal{K} = -\frac{w_t}{x_t^2} \mathbf{x}_s + \frac{1}{x_t} \mathbf{w}_s = -\frac{\mathbf{x}_s \cdot \mathbf{v}_s}{x_t^2} \mathbf{x}_s + \frac{1}{x_t} \left(\mathbf{v}_s - \frac{\mathbf{x}_s \cdot \mathbf{v}_s}{(x_t - 1)} \mathbf{x}_s \right). \quad (60)$$

Expressing $x_t, \mathbf{x}_s$ in terms of $\mathbf{x}^\mathcal{K}$, we have

$$x_t = \frac{1}{\sqrt{1 - \|\mathbf{x}^\mathcal{K}\|^2}}, \quad \mathbf{x}_s = \frac{1}{\sqrt{1 - \|\mathbf{x}^\mathcal{K}\|^2}} \mathbf{x}^\mathcal{K}. \quad (61)$$

We further express $\mathbf{v}^\mathcal{L} \in \mathcal{T}_{\mathbf{o}^\mathcal{L}} \mathbb{L}^n$ in terms of the corresponding tangent vector $\mathbf{v}^\mathcal{K}$ in $\mathcal{T}_{\mathbf{o}^\mathcal{K}} \mathbb{K}^n$. We already know that $v_t = 0$. Also,

$$\mathbf{v}_s = \frac{1}{\sqrt{1 - \|\mathbf{o}^\mathcal{K}\|^2}} \mathbf{v}^\mathcal{K} + \frac{\mathbf{o}^\mathcal{K} \cdot \mathbf{v}^\mathcal{K}}{(1 - \|\mathbf{o}^\mathcal{K}\|^2)^{\frac{3}{2}}} \mathbf{o}^\mathcal{K} = \mathbf{v}^\mathcal{K}. \quad (62)$$

Hence,

$$\begin{aligned}
 P_{\mathbf{o}^{\mathcal{L}} \rightarrow \mathbf{x}^{\mathcal{L}}}(\mathbf{v}^{\mathcal{L}}) &= \mathbf{w}^{\mathcal{K}} = -\frac{\mathbf{x}_s \cdot \mathbf{v}_s}{x_t^2} \mathbf{x}_s + \frac{1}{x_t} \left(\mathbf{v}_s - \frac{\mathbf{x}_s \cdot \mathbf{v}_s}{(x_t - 1)} \mathbf{x}_s \right) \\
 &= -(1 - \|\mathbf{x}^{\mathcal{K}}\|^2) \frac{\mathbf{x}^{\mathcal{K}} \cdot \mathbf{v}^{\mathcal{K}}}{\sqrt{1 - \|\mathbf{x}^{\mathcal{K}}\|^2}} \frac{1}{\sqrt{1 - \|\mathbf{x}^{\mathcal{K}}\|^2}} \mathbf{x}^{\mathcal{K}} \\
 &\quad + \sqrt{1 - \|\mathbf{x}^{\mathcal{K}}\|^2} \left(\mathbf{v}^{\mathcal{K}} - \frac{1}{\frac{1}{\sqrt{1 - \|\mathbf{x}^{\mathcal{K}}\|^2}} - 1} \frac{\mathbf{x}^{\mathcal{K}} \cdot \mathbf{v}^{\mathcal{K}}}{\sqrt{1 - \|\mathbf{x}^{\mathcal{K}}\|^2}} \frac{1}{\sqrt{1 - \|\mathbf{x}^{\mathcal{K}}\|^2}} \mathbf{x}^{\mathcal{K}} \right) \\
 &= -(\mathbf{x}^{\mathcal{K}} \cdot \mathbf{v}^{\mathcal{K}}) \mathbf{x}^{\mathcal{K}} + \sqrt{1 - \|\mathbf{x}^{\mathcal{K}}\|^2} \mathbf{v}^{\mathcal{K}} - \frac{\mathbf{x}^{\mathcal{K}} \cdot \mathbf{v}^{\mathcal{K}}}{1 - \sqrt{1 - \|\mathbf{x}^{\mathcal{K}}\|^2}} \mathbf{x}^{\mathcal{K}} \\
 &= \frac{(\mathbf{x}^{\mathcal{K}} \cdot \mathbf{v}^{\mathcal{K}})(\sqrt{1 - \|\mathbf{x}^{\mathcal{K}}\|^2} - 2)}{1 - \sqrt{1 - \|\mathbf{x}^{\mathcal{K}}\|^2}} \mathbf{x}^{\mathcal{K}} + \sqrt{1 - \|\mathbf{x}^{\mathcal{K}}\|^2} \mathbf{v}^{\mathcal{K}}. \tag{63}
 \end{aligned}$$

■

B.4. Proof of Theorem 6

Recall that Corollary 4 indicates

$$\exp_{\mathbf{o}^{\mathcal{K}}}^{\mathcal{K}}(\mathbf{v}^{\mathcal{K}}) = \tanh(\|\mathbf{v}^{\mathcal{K}}\|) \frac{\mathbf{v}^{\mathcal{K}}}{\|\mathbf{v}^{\mathcal{K}}\|}, \tag{64}$$

$$\log_{\mathbf{o}^{\mathcal{K}}}^{\mathcal{K}}(\mathbf{x}^{\mathcal{K}}) = \cosh^{-1} \left(\frac{1}{\sqrt{1 - \|\mathbf{x}^{\mathcal{K}}\|^2}} \right) \frac{\mathbf{x}^{\mathcal{K}}}{\|\mathbf{x}^{\mathcal{K}}\|}. \tag{65}$$

On the one hand,

$$\begin{aligned}
 \exp_{\mathbf{o}^{\mathcal{K}}}^{\mathcal{K}}(r \log_{\mathbf{o}^{\mathcal{K}}}^{\mathcal{K}}(\mathbf{x}^{\mathcal{K}})) &= \exp_{\mathbf{o}^{\mathcal{K}}}^{\mathcal{K}} \left(r \cosh^{-1} \left(\frac{1}{\sqrt{1 - \|\mathbf{x}^{\mathcal{K}}\|^2}} \right) \frac{\mathbf{x}^{\mathcal{K}}}{\|\mathbf{x}^{\mathcal{K}}\|} \right) \\
 &= \tanh \left(|r| \cosh^{-1} \left(\frac{1}{\sqrt{1 - \|\mathbf{x}^{\mathcal{K}}\|^2}} \right) \right) \frac{r \mathbf{x}^{\mathcal{K}}}{|r| \|\mathbf{x}^{\mathcal{K}}\|} \\
 &= \tanh \left(r \cosh^{-1} \left(\frac{1}{\sqrt{1 - \|\mathbf{x}^{\mathcal{K}}\|^2}} \right) \right) \frac{\mathbf{x}^{\mathcal{K}}}{\|\mathbf{x}^{\mathcal{K}}\|}. \tag{66}
 \end{aligned}$$

On the other hand,

$$r \otimes_{\mathbf{E}} \mathbf{x}^{\mathcal{K}} = \tanh(r \tanh^{-1}(\|\mathbf{x}^{\mathcal{K}}\|)) \frac{\mathbf{x}^{\mathcal{K}}}{\|\mathbf{x}^{\mathcal{K}}\|}. \tag{67}$$

Therefore, to prove $r \otimes_{\mathbf{E}} \mathbf{x}^{\mathcal{K}} = \exp_{\mathbf{o}^{\mathcal{K}}}^{\mathcal{K}}(r \log_{\mathbf{o}^{\mathcal{K}}}^{\mathcal{K}}(\mathbf{x}^{\mathcal{K}}))$, it suffices to show:

$$\cosh^{-1} \left(\frac{1}{\sqrt{1 - \|\mathbf{x}^{\mathcal{K}}\|^2}} \right) = \tanh^{-1}(\|\mathbf{x}^{\mathcal{K}}\|). \tag{68}$$

Let $\cosh^{-1}\left(\frac{1}{\sqrt{1-\|\mathbf{x}^{\mathcal{K}}\|^2}}\right) = t$, then $\cosh(t) = \frac{1}{\sqrt{1-\|\mathbf{x}^{\mathcal{K}}\|^2}}$, i.e.,

$$\frac{e^t + e^{-t}}{2} = \frac{1}{\sqrt{1-\|\mathbf{x}^{\mathcal{K}}\|^2}}. \quad (69)$$

This implies that

$$\|\mathbf{x}^{\mathcal{K}}\|^2 = 1 - \frac{4}{e^{2t} + e^{-2t} + 2} = \frac{e^{2t} + e^{-2t} - 2}{e^{2t} + e^{-2t} + 2} = \left(\frac{e^t - e^{-t}}{e^t + e^{-t}}\right)^2. \quad (70)$$

Therefore, $\|\mathbf{x}^{\mathcal{K}}\| = \frac{e^t - e^{-t}}{e^t + e^{-t}}$ since $t > 0$, i.e. $\tanh^{-1}(\|\mathbf{x}^{\mathcal{K}}\|) = t$, completing the proof. $\blacksquare$

B.5. Proof of Theorem 7

Our proof is based on Equation (28) and the isometric mappings in Equations (3)–(5).

Note that proving Equation (18) is equivalent to showing that $P_{\mathbf{o}^{\mathcal{K}} \rightarrow \mathbf{x}^{\mathcal{K}}}(\mathbf{v}^{\mathcal{K}})$ given by $\log_{\mathbf{x}^{\mathcal{K}}}^{\mathcal{K}}(\mathbf{x}^{\mathcal{K}} \oplus_{\mathbf{E}} \exp_{\mathbf{o}^{\mathcal{K}}}^{\mathcal{K}}(\mathbf{v}^{\mathcal{K}}))$ and $P_{\mathbf{o}^{\mathcal{B}} \rightarrow \mathbf{x}^{\mathcal{B}}}(\mathbf{v}^{\mathcal{B}})$ given by $\log_{\mathbf{x}^{\mathcal{B}}}^{\mathcal{B}}(\mathbf{x}^{\mathcal{B}} \oplus_{\mathbf{M}} \exp_{\mathbf{o}^{\mathcal{B}}}^{\mathcal{B}}(\mathbf{v}^{\mathcal{B}}))$ are corresponding tangent vectors. This is further equivalent to showing that $\mathbf{x}^{\mathcal{K}} \oplus_{\mathbf{E}} \exp_{\mathbf{o}^{\mathcal{K}}}^{\mathcal{K}}(\mathbf{v}^{\mathcal{K}})$ and $\mathbf{x}^{\mathcal{B}} \oplus_{\mathbf{M}} \exp_{\mathbf{o}^{\mathcal{B}}}^{\mathcal{B}}(\mathbf{v}^{\mathcal{B}})$ are corresponding points in hyperbolic space.

On the one hand, applying Equation (26), we have

$$\begin{aligned} \mathbf{x}^{\mathcal{B}} \oplus_{\mathbf{M}} \exp_{\mathbf{o}^{\mathcal{B}}}^{\mathcal{B}}(\mathbf{v}^{\mathcal{B}}) &= \mathbf{x}^{\mathcal{B}} \oplus_{\mathbf{M}} \left(\tanh(\|\mathbf{v}^{\mathcal{B}}\|) \frac{\mathbf{v}^{\mathcal{B}}}{\|\mathbf{v}^{\mathcal{B}}\|} \right) \\ &= \frac{\left(1 + \frac{2 \tanh(\|\mathbf{v}^{\mathcal{B}}\|)}{\|\mathbf{v}^{\mathcal{B}}\|} \mathbf{x}^{\mathcal{B}} \cdot \mathbf{v}^{\mathcal{B}} + \tanh^2(\|\mathbf{v}^{\mathcal{B}}\|) \right) \mathbf{x}^{\mathcal{B}} + (1 - \|\mathbf{x}^{\mathcal{B}}\|^2) \frac{\tanh(\|\mathbf{v}^{\mathcal{B}}\|)}{\|\mathbf{v}^{\mathcal{B}}\|} \mathbf{v}^{\mathcal{B}}}{1 + \frac{2 \tanh(\|\mathbf{v}^{\mathcal{B}}\|)}{\|\mathbf{v}^{\mathcal{B}}\|} \mathbf{x}^{\mathcal{B}} \cdot \mathbf{v}^{\mathcal{B}} + \tanh^2(\|\mathbf{v}^{\mathcal{B}}\|) \|\mathbf{x}^{\mathcal{B}}\|^2} \\ &= \frac{(\|\mathbf{v}^{\mathcal{B}}\| + 2 \tanh(\|\mathbf{v}^{\mathcal{B}}\|) \mathbf{x}^{\mathcal{B}} \cdot \mathbf{v}^{\mathcal{B}} + \tanh^2(\|\mathbf{v}^{\mathcal{B}}\|) \|\mathbf{v}^{\mathcal{B}}\|) \mathbf{x}^{\mathcal{B}} + (1 - \|\mathbf{x}^{\mathcal{B}}\|^2) \tanh(\|\mathbf{v}^{\mathcal{B}}\|) \mathbf{v}^{\mathcal{B}}}{\|\mathbf{v}^{\mathcal{B}}\| + 2 \tanh(\|\mathbf{v}^{\mathcal{B}}\|) \mathbf{x}^{\mathcal{B}} \cdot \mathbf{v}^{\mathcal{B}} + \tanh^2(\|\mathbf{v}^{\mathcal{B}}\|) \|\mathbf{x}^{\mathcal{B}}\|^2 \|\mathbf{v}^{\mathcal{B}}\|}. \end{aligned} \quad (71)$$

On the other hand, applying Equation (14), we have

$$\begin{aligned} \mathbf{x}^{\mathcal{K}} \oplus_{\mathbf{E}} \exp_{\mathbf{o}^{\mathcal{K}}}^{\mathcal{K}}(\mathbf{v}^{\mathcal{K}}) &= \mathbf{x}^{\mathcal{K}} \oplus_{\mathbf{E}} \left(\tanh(\|\mathbf{v}^{\mathcal{K}}\|) \frac{\mathbf{v}^{\mathcal{K}}}{\|\mathbf{v}^{\mathcal{K}}\|} \right) \\ &= \frac{1}{1 + \frac{\tanh(\|\mathbf{v}^{\mathcal{K}}\|)}{\|\mathbf{v}^{\mathcal{K}}\|} \mathbf{x}^{\mathcal{K}} \cdot \mathbf{v}^{\mathcal{K}}} \left(\mathbf{x}^{\mathcal{K}} + \sqrt{1 - \|\mathbf{x}^{\mathcal{K}}\|^2} \frac{\tanh(\|\mathbf{v}^{\mathcal{K}}\|)}{\|\mathbf{v}^{\mathcal{K}}\|} \mathbf{v}^{\mathcal{K}} \right. \\ &\quad \left. + \frac{1}{1 + \frac{1}{\sqrt{1 - \|\mathbf{x}^{\mathcal{K}}\|^2}}} \frac{\tanh(\|\mathbf{v}^{\mathcal{K}}\|)}{\|\mathbf{v}^{\mathcal{K}}\|} (\mathbf{x}^{\mathcal{K}} \cdot \mathbf{v}^{\mathcal{K}}) \mathbf{x}^{\mathcal{K}} \right) \\ &= \frac{\|\mathbf{v}^{\mathcal{K}}\|}{\|\mathbf{v}^{\mathcal{K}}\| + \tanh(\|\mathbf{v}^{\mathcal{K}}\|) \mathbf{x}^{\mathcal{K}} \cdot \mathbf{v}^{\mathcal{K}}} \left(\mathbf{x}^{\mathcal{K}} + \sqrt{1 - \|\mathbf{x}^{\mathcal{K}}\|^2} \frac{\tanh(\|\mathbf{v}^{\mathcal{K}}\|)}{\|\mathbf{v}^{\mathcal{K}}\|} \mathbf{v}^{\mathcal{K}} \right. \\ &\quad \left. + \frac{1}{1 + \sqrt{1 - \|\mathbf{x}^{\mathcal{K}}\|^2}} \frac{\tanh(\|\mathbf{v}^{\mathcal{K}}\|)}{\|\mathbf{v}^{\mathcal{K}}\|} (\mathbf{x}^{\mathcal{K}} \cdot \mathbf{v}^{\mathcal{K}}) \mathbf{x}^{\mathcal{K}} \right) \end{aligned}$$

$$\begin{aligned}
 &= \frac{\|\mathbf{v}^{\mathcal{K}}\|}{\|\mathbf{v}^{\mathcal{K}}\| + \tanh(\|\mathbf{v}^{\mathcal{K}}\|)\mathbf{x}^{\mathcal{K}} \cdot \mathbf{v}^{\mathcal{K}}} \mathbf{x}^{\mathcal{K}} + \frac{\tanh(\|\mathbf{v}^{\mathcal{K}}\|)\mathbf{x}^{\mathcal{K}} \cdot \mathbf{v}^{\mathcal{K}}}{\|\mathbf{v}^{\mathcal{K}}\| + \tanh(\|\mathbf{v}^{\mathcal{K}}\|)\mathbf{x}^{\mathcal{K}} \cdot \mathbf{v}^{\mathcal{K}}} \frac{\mathbf{x}^{\mathcal{K}}}{1 + \sqrt{1 - \|\mathbf{x}^{\mathcal{K}}\|^2}} \\
 &\quad + \frac{\tanh(\|\mathbf{v}^{\mathcal{K}}\|)\sqrt{1 - \|\mathbf{x}^{\mathcal{K}}\|^2}}{\|\mathbf{v}^{\mathcal{K}}\| + \tanh(\|\mathbf{v}^{\mathcal{K}}\|)\mathbf{x}^{\mathcal{K}} \cdot \mathbf{v}^{\mathcal{K}}} \mathbf{v}^{\mathcal{K}}.
 \end{aligned}$$

Given $\mathbf{x}^{\mathcal{B}}, \mathbf{v}^{\mathcal{B}}$, the corresponding $\mathbf{x}^{\mathcal{K}} \in \mathbb{K}^n, \mathbf{v}^{\mathcal{K}} \in \mathcal{T}_{\mathbf{o}^{\mathcal{K}}} \mathbb{K}^n$ are

$$\mathbf{x}^{\mathcal{K}} = \frac{2}{1 + \|\mathbf{x}^{\mathcal{B}}\|^2} \mathbf{x}^{\mathcal{B}}, \quad \mathbf{v}^{\mathcal{K}} = \frac{2}{1 + \|\mathbf{o}^{\mathcal{B}}\|^2} \mathbf{v}^{\mathcal{B}} - \frac{4\mathbf{o}^{\mathcal{B}} \cdot \mathbf{v}^{\mathcal{B}}}{(1 + \|\mathbf{o}^{\mathcal{B}}\|^2)^2} \mathbf{o}^{\mathcal{B}} = 2\mathbf{v}^{\mathcal{B}}. \quad (72)$$

Writing $\mathbf{x}^{\mathcal{K}} \oplus_{\mathbb{E}} \exp_{\mathbf{o}^{\mathcal{K}}}^{\mathcal{K}}(\mathbf{v}^{\mathcal{K}})$ in terms of $\mathbf{x}^{\mathcal{B}}, \mathbf{v}^{\mathcal{B}}$ yields

$$\begin{aligned}
 &\mathbf{x}^{\mathcal{K}} \oplus_{\mathbb{E}} \exp_{\mathbf{o}^{\mathcal{K}}}^{\mathcal{K}}(\mathbf{v}^{\mathcal{K}}) \\
 &= \frac{2\|\mathbf{v}^{\mathcal{B}}\|}{2\|\mathbf{v}^{\mathcal{B}}\| + \tanh(2\|\mathbf{v}^{\mathcal{B}}\|)\frac{2\mathbf{x}^{\mathcal{B}} \cdot 2\mathbf{v}^{\mathcal{B}}}{1 + \|\mathbf{x}^{\mathcal{B}}\|^2}} \frac{2\mathbf{x}^{\mathcal{B}}}{1 + \|\mathbf{x}^{\mathcal{B}}\|^2} + \frac{\tanh(2\|\mathbf{v}^{\mathcal{B}}\|)\frac{2\mathbf{x}^{\mathcal{B}} \cdot 2\mathbf{v}^{\mathcal{B}}}{1 + \|\mathbf{x}^{\mathcal{B}}\|^2}}{2\|\mathbf{v}^{\mathcal{B}}\| + \tanh(2\|\mathbf{v}^{\mathcal{B}}\|)\frac{2\mathbf{x}^{\mathcal{B}} \cdot 2\mathbf{v}^{\mathcal{B}}}{1 + \|\mathbf{x}^{\mathcal{B}}\|^2}} \mathbf{x}^{\mathcal{B}} \\
 &\quad + \frac{\tanh(2\|\mathbf{v}^{\mathcal{B}}\|)\sqrt{1 - \frac{4\|\mathbf{x}^{\mathcal{B}}\|^2}{(1 + \|\mathbf{x}^{\mathcal{B}}\|^2)^2}}}{2\|\mathbf{v}^{\mathcal{B}}\| + \tanh(2\|\mathbf{v}^{\mathcal{B}}\|)\frac{2\mathbf{x}^{\mathcal{B}} \cdot 2\mathbf{v}^{\mathcal{B}}}{1 + \|\mathbf{x}^{\mathcal{B}}\|^2}} 2\mathbf{v}^{\mathcal{B}} \\
 &= \frac{\|\mathbf{v}^{\mathcal{B}}\| (1 + \|\mathbf{x}^{\mathcal{B}}\|^2)}{\|\mathbf{v}^{\mathcal{B}}\| (1 + \|\mathbf{x}^{\mathcal{B}}\|^2) + 2 \tanh(2\|\mathbf{v}^{\mathcal{B}}\|)\mathbf{x}^{\mathcal{B}} \cdot \mathbf{v}^{\mathcal{B}}} \frac{2\mathbf{x}^{\mathcal{B}}}{1 + \|\mathbf{x}^{\mathcal{B}}\|^2} \\
 &\quad + \frac{2 \tanh(2\|\mathbf{v}^{\mathcal{B}}\|)\mathbf{x}^{\mathcal{B}} \cdot \mathbf{v}^{\mathcal{B}}}{\|\mathbf{v}^{\mathcal{B}}\| (1 + \|\mathbf{x}^{\mathcal{B}}\|^2) + 2 \tanh(2\|\mathbf{v}^{\mathcal{B}}\|)\mathbf{x}^{\mathcal{B}} \cdot \mathbf{v}^{\mathcal{B}}} \mathbf{x}^{\mathcal{B}} \\
 &\quad + \frac{\tanh(2\|\mathbf{v}^{\mathcal{B}}\|) (1 - \|\mathbf{x}^{\mathcal{B}}\|^2)}{\|\mathbf{v}^{\mathcal{B}}\| (1 + \|\mathbf{x}^{\mathcal{B}}\|^2) + 2 \tanh(2\|\mathbf{v}^{\mathcal{B}}\|)\mathbf{x}^{\mathcal{B}} \cdot \mathbf{v}^{\mathcal{B}}} \mathbf{v}^{\mathcal{B}} \\
 &= \frac{2\|\mathbf{v}^{\mathcal{B}}\| + 2 \tanh(2\|\mathbf{v}^{\mathcal{B}}\|)\mathbf{x}^{\mathcal{B}} \cdot \mathbf{v}^{\mathcal{B}}}{\|\mathbf{v}^{\mathcal{B}}\| (1 + \|\mathbf{x}^{\mathcal{B}}\|^2) + 2 \tanh(2\|\mathbf{v}^{\mathcal{B}}\|)\mathbf{x}^{\mathcal{B}} \cdot \mathbf{v}^{\mathcal{B}}} \mathbf{x}^{\mathcal{B}} \\
 &\quad + \frac{\tanh(2\|\mathbf{v}^{\mathcal{B}}\|) (1 - \|\mathbf{x}^{\mathcal{B}}\|^2)}{\|\mathbf{v}^{\mathcal{B}}\| (1 + \|\mathbf{x}^{\mathcal{B}}\|^2) + 2 \tanh(2\|\mathbf{v}^{\mathcal{B}}\|)\mathbf{x}^{\mathcal{B}} \cdot \mathbf{v}^{\mathcal{B}}} \mathbf{v}^{\mathcal{B}} \\
 &= \frac{2\|\mathbf{v}^{\mathcal{B}}\| + \frac{4 \tanh(\|\mathbf{v}^{\mathcal{B}}\|)\mathbf{x}^{\mathcal{B}} \cdot \mathbf{v}^{\mathcal{B}}}{1 + \tanh^2(\|\mathbf{v}^{\mathcal{B}}\|)}}{\|\mathbf{v}^{\mathcal{B}}\| (1 + \|\mathbf{x}^{\mathcal{B}}\|^2) + \frac{4 \tanh(\|\mathbf{v}^{\mathcal{B}}\|)\mathbf{x}^{\mathcal{B}} \cdot \mathbf{v}^{\mathcal{B}}}{1 + \tanh^2(\|\mathbf{v}^{\mathcal{B}}\|)}} \mathbf{x}^{\mathcal{B}} + \frac{\frac{2 \tanh(\|\mathbf{v}^{\mathcal{B}}\|)(1 - \|\mathbf{x}^{\mathcal{B}}\|^2)}{1 + \tanh^2(\|\mathbf{v}^{\mathcal{B}}\|)}}{\|\mathbf{v}^{\mathcal{B}}\| (1 + \|\mathbf{x}^{\mathcal{B}}\|^2) + \frac{4 \tanh(\|\mathbf{v}^{\mathcal{B}}\|)\mathbf{x}^{\mathcal{B}} \cdot \mathbf{v}^{\mathcal{B}}}{1 + \tanh^2(\|\mathbf{v}^{\mathcal{B}}\|)}} \mathbf{v}^{\mathcal{B}} \\
 &= \frac{2\|\mathbf{v}^{\mathcal{B}}\| (1 + \tanh^2(\|\mathbf{v}^{\mathcal{B}}\|)) + 4 \tanh(\|\mathbf{v}^{\mathcal{B}}\|)\mathbf{x}^{\mathcal{B}} \cdot \mathbf{v}^{\mathcal{B}}}{\|\mathbf{v}^{\mathcal{B}}\| (1 + \|\mathbf{x}^{\mathcal{B}}\|^2) (1 + \tanh^2(\|\mathbf{v}^{\mathcal{B}}\|)) + 4 \tanh(\|\mathbf{v}^{\mathcal{B}}\|)\mathbf{x}^{\mathcal{B}} \cdot \mathbf{v}^{\mathcal{B}}} \mathbf{x}^{\mathcal{B}} \\
 &\quad + \frac{2 \tanh(\|\mathbf{v}^{\mathcal{B}}\|) (1 - \|\mathbf{x}^{\mathcal{B}}\|^2)}{\|\mathbf{v}^{\mathcal{B}}\| (1 + \|\mathbf{x}^{\mathcal{B}}\|^2) (1 + \tanh^2(\|\mathbf{v}^{\mathcal{B}}\|)) + 4 \tanh(\|\mathbf{v}^{\mathcal{B}}\|)\mathbf{x}^{\mathcal{B}} \cdot \mathbf{v}^{\mathcal{B}}} \mathbf{v}^{\mathcal{B}} \\
 &= \frac{(2\|\mathbf{v}^{\mathcal{B}}\| + 2 \tanh^2(\|\mathbf{v}^{\mathcal{B}}\|)\|\mathbf{v}^{\mathcal{B}}\| + 4 \tanh(\|\mathbf{v}^{\mathcal{B}}\|)\mathbf{x}^{\mathcal{B}} \cdot \mathbf{v}^{\mathcal{B}}) \mathbf{x}^{\mathcal{B}} + (2 \tanh(\|\mathbf{v}^{\mathcal{B}}\| - 2 \tanh(\|\mathbf{v}^{\mathcal{B}}\|)\|\mathbf{x}^{\mathcal{B}}\|^2) \mathbf{v}^{\mathcal{B}}}{\|\mathbf{v}^{\mathcal{B}}\| + \|\mathbf{x}^{\mathcal{B}}\|^2\|\mathbf{v}^{\mathcal{B}}\| + \tanh^2(\|\mathbf{v}^{\mathcal{B}}\|)\|\mathbf{v}^{\mathcal{B}}\| + \tanh^2(\|\mathbf{v}^{\mathcal{B}}\|)\|\mathbf{x}^{\mathcal{B}}\|^2\|\mathbf{v}^{\mathcal{B}}\| + 4 \tanh(\|\mathbf{v}^{\mathcal{B}}\|)\mathbf{x}^{\mathcal{B}} \cdot \mathbf{v}^{\mathcal{B}}}
 \end{aligned}$$

Meanwhile, we need to map $\mathbf{x}^{\mathcal{B}} \oplus_{\mathbb{M}} \exp_{\mathbf{o}^{\mathcal{B}}}^{\mathcal{B}}(\mathbf{v}^{\mathcal{B}}) \in \mathbb{B}^n$ to $\mathbb{K}^n$ and check whether it matches the simplified result of $\mathbf{x}^{\mathcal{K}} \oplus_{\mathbb{E}} \exp_{\mathbf{o}^{\mathcal{K}}}^{\mathcal{K}}(\mathbf{v}^{\mathcal{K}})$. Denote

$$\|\mathbf{v}^{\mathcal{B}}\| + 2 \tanh(\|\mathbf{v}^{\mathcal{B}}\|)\mathbf{x}^{\mathcal{B}} \cdot \mathbf{v}^{\mathcal{B}} + \tanh^2(\|\mathbf{v}^{\mathcal{B}}\|)\|\mathbf{x}^{\mathcal{B}}\|^2\|\mathbf{v}^{\mathcal{B}}\| = A, \quad (73)$$

$$\|\mathbf{v}^{\mathcal{B}}\| + 2 \tanh(\|\mathbf{v}^{\mathcal{B}}\|) \mathbf{x}^{\mathcal{B}} \cdot \mathbf{v}^{\mathcal{B}} + \tanh^2(\|\mathbf{v}^{\mathcal{B}}\|) \|\mathbf{v}^{\mathcal{B}}\| = B, \quad (74)$$

$$(1 - \|\mathbf{x}^{\mathcal{B}}\|^2) \tanh(\|\mathbf{v}^{\mathcal{B}}\|) = C. \quad (75)$$

Then $\mathbf{x}^{\mathcal{B}} \oplus_{\mathbb{M}} \exp_{\mathbf{o}^{\mathcal{B}}}^{\mathcal{B}}(\mathbf{v}^{\mathcal{B}}) \in \mathbb{B}^n$ can be written as $\frac{B\mathbf{x}^{\mathcal{B}} + C\mathbf{v}^{\mathcal{B}}}{A}$. Thus, the corresponding point in $\mathbb{K}^n$ is given by

$$\begin{aligned} \frac{2 \frac{B\mathbf{x}^{\mathcal{B}} + C\mathbf{v}^{\mathcal{B}}}{A}}{1 + \left\| \frac{B\mathbf{x}^{\mathcal{B}} + C\mathbf{v}^{\mathcal{B}}}{A} \right\|^2} &= \frac{2AB\mathbf{x}^{\mathcal{B}} + 2AC\mathbf{v}^{\mathcal{B}}}{A^2 + \|B\mathbf{x}^{\mathcal{B}} + C\mathbf{v}^{\mathcal{B}}\|^2} \\ &= \frac{2AB\mathbf{x}^{\mathcal{B}} + 2AC\mathbf{v}^{\mathcal{B}}}{A^2 + B^2\|\mathbf{x}^{\mathcal{B}}\|^2 + C^2\|\mathbf{v}^{\mathcal{B}}\|^2 + 2BC(\mathbf{x}^{\mathcal{B}} \cdot \mathbf{v}^{\mathcal{B}})}. \end{aligned} \quad (76)$$

For clarity, let $\|\mathbf{x}^{\mathcal{B}}\| = x$, $\|\mathbf{v}^{\mathcal{B}}\| = v$, and $\mathbf{x}^{\mathcal{B}} \cdot \mathbf{v}^{\mathcal{B}} = xv \cos \theta$. We calculate:

$$\begin{aligned} A^2 &= (v + 2xv \cos(\theta) \tanh(v) + x^2v \tanh^2(v))^2 \\ &= v^2 + 4xv^2 \cos(\theta) \tanh(v) + 2x^2v^2 \tanh^2(v) \\ &\quad + 4x^2v^2 \cos^2(\theta) \tanh^2(v) + 4x^3v^2 \cos(\theta) \tanh^3(v) + x^4v^2 \tanh^4(v), \end{aligned} \quad (77)$$

$$\begin{aligned} B^2\|\mathbf{x}^{\mathcal{B}}\|^2 &= (v + 2xv \cos(\theta) \tanh(v) + v \tanh^2(v))^2 x^2 \\ &= x^2v^2 + 4x^3v^2 \cos(\theta) \tanh(v) + 2x^2v^2 \tanh^2(v) \\ &\quad + 4x^4v^2 \cos^2(\theta) \tanh^2(v) + 4x^3v^2 \cos(\theta) \tanh^3(v) + x^2v^2 \tanh^4(v), \end{aligned} \quad (78)$$

$$C^2\|\mathbf{v}^{\mathcal{B}}\|^2 = (1 - x^2)^2 v^2 \tanh^2(v) = v^2 \tanh^2(v) - 2x^2v^2 \tanh^2(v) + x^4v^2 \tanh^2(v), \quad (79)$$

$$\begin{aligned} 2BC(\mathbf{x}^{\mathcal{B}} \cdot \mathbf{v}^{\mathcal{B}}) &= 2xv \cos(\theta) (v + 2xv \cos(\theta) \tanh(v) + v \tanh^2(v)) (1 - x^2) \tanh(v) \\ &= 2xv^2 \cos(\theta) \tanh(v) - 2x^3v^2 \cos(\theta) \tanh(v) + 4x^2v^2 \cos^2(\theta) \tanh^2(v) \\ &\quad - 4x^4v^2 \cos^2(\theta) \tanh^2(v) + 2xv^2 \cos(\theta) \tanh^3(v) - 2x^3v^2 \cos(\theta) \tanh^3(v). \end{aligned} \quad (80)$$

Thus, the denominator in Equation (76) is

$$\begin{aligned} &A^2 + B^2\|\mathbf{x}^{\mathcal{B}}\|^2 + C^2\|\mathbf{v}^{\mathcal{B}}\|^2 + 2BC(\mathbf{x}^{\mathcal{B}} \cdot \mathbf{v}^{\mathcal{B}}) \\ &= v^2 + x^2v^2 + 6xv^2 \cos(\theta) \tanh(v) + 2x^3v^2 \cos(\theta) \tanh(v) \\ &\quad + v^2 \tanh^2(v) + 2x^2v^2 \tanh^2(v) + x^4v^2 \tanh^2(v) + 8x^2v^2 \cos^2(\theta) \tanh^2(v) \\ &\quad + 2xv^2 \cos(\theta) \tanh^3(v) + 6x^3v^2 \cos(\theta) \tanh^3(v) + x^2v^2 \tanh^4(v) + x^4v^2 \tanh^4(v). \end{aligned} \quad (81)$$

We can factor this expression and get

$$\begin{aligned} &A^2 + B^2\|\mathbf{x}^{\mathcal{B}}\|^2 + C^2\|\mathbf{v}^{\mathcal{B}}\|^2 + 2BC(\mathbf{x}^{\mathcal{B}} \cdot \mathbf{v}^{\mathcal{B}}) \\ &= v^2 (1 + 2x \cos(\theta) \tanh(v) + x^2 \tanh^2(v)) \\ &\quad \cdot (1 + x^2 + 4x \cos(\theta) \tanh(v) + \tanh^2(v) + x^2 \tanh^2(v)). \end{aligned} \quad (82)$$

The numerator contains the following terms:

$$2AB = 2 (v + 2xv \cos(\theta) \tanh(v) + x^2v \tanh^2(v)) (v + 2xv \cos(\theta) \tanh(v) + v \tanh^2(v))$$

$$= 2v^2 (1 + 2x \cos(\theta) \tanh(v) + x^2 \tanh^2(v)) (1 + 2x \cos(\theta) \tanh(v) + \tanh^2(v)), \quad (83)$$

$$\begin{aligned} 2AC &= 2 (v + 2xv \cos(\theta) \tanh(v) + x^2 v \tanh^2(v)) (1 - x^2) \tanh(v) \\ &= 2v (1 + 2x \cos(\theta) \tanh(v) + x^2 \tanh^2(v)) (\tanh(v) - x^2 \tanh(v)). \end{aligned} \quad (84)$$

Altogether, Equation (76) can be simplified as

$$\begin{aligned} & \frac{2AB\mathbf{x}^{\mathcal{B}} + 2AC\mathbf{v}^{\mathcal{B}}}{A^2 + B^2\|\mathbf{x}^{\mathcal{B}}\|^2 + C^2\|\mathbf{v}^{\mathcal{B}}\|^2 + 2BC(\mathbf{x}^{\mathcal{B}} \cdot \mathbf{v}^{\mathcal{B}})} \\ &= \frac{2v^2 (1 + 2x \cos(\theta) \tanh(v) + x^2 \tanh^2(v)) (1 + 2x \cos(\theta) \tanh(v) + \tanh^2(v))}{v^2 (1 + 2x \cos(\theta) \tanh(v) + x^2 \tanh^2(v)) (1 + x^2 + 4x \cos(\theta) \tanh(v) + \tanh^2(v) + x^2 \tanh^2(v))} \mathbf{x}^{\mathcal{B}} \\ &+ \frac{2v (1 + 2x \cos(\theta) \tanh(v) + x^2 \tanh^2(v)) (\tanh(v) - x^2 \tanh(v))}{v^2 (1 + 2x \cos(\theta) \tanh(v) + x^2 \tanh^2(v)) (1 + x^2 + 4x \cos(\theta) \tanh(v) + \tanh^2(v) + x^2 \tanh^2(v))} \mathbf{v}^{\mathcal{B}} \\ &= \frac{2 (1 + 2x \cos(\theta) \tanh(v) + \tanh^2(v))}{1 + x^2 + 4x \cos(\theta) \tanh(v) + \tanh^2(v) + x^2 \tanh^2(v)} \mathbf{x}^{\mathcal{B}} \\ &+ \frac{2 (\tanh(v) - x^2 \tanh(v))}{v (1 + x^2 + 4x \cos(\theta) \tanh(v) + \tanh^2(v) + x^2 \tanh^2(v))} \mathbf{v}^{\mathcal{B}}. \end{aligned} \quad (85)$$

If we express $\mathbf{x}^{\mathcal{K}} \oplus_{\mathbf{E}} \exp_{\mathbf{o}^{\mathcal{K}}}^{\mathcal{K}}(\mathbf{v}^{\mathcal{K}})$ in terms of x, v and $\cos(\theta)$, we get

$$\begin{aligned} & \mathbf{x}^{\mathcal{K}} \oplus_{\mathbf{E}} \exp_{\mathbf{o}^{\mathcal{K}}}^{\mathcal{K}}(\mathbf{v}^{\mathcal{K}}) \\ &= \frac{(2\|\mathbf{v}^{\mathcal{B}}\| + 2 \tanh^2(\|\mathbf{v}^{\mathcal{B}}\|)\|\mathbf{v}^{\mathcal{B}}\| + 4 \tanh(\|\mathbf{v}^{\mathcal{B}}\|)\mathbf{x}^{\mathcal{B}} \cdot \mathbf{v}^{\mathcal{B}}) \mathbf{x}^{\mathcal{B}} + (2 \tanh(\|\mathbf{v}^{\mathcal{B}}\| - 2 \tanh(\|\mathbf{v}^{\mathcal{B}}\|)\|\mathbf{x}^{\mathcal{B}}\|^2) \mathbf{v}^{\mathcal{B}}}{\|\mathbf{v}^{\mathcal{B}}\| + \|\mathbf{x}^{\mathcal{B}}\|^2\|\mathbf{v}^{\mathcal{B}}\| + \tanh^2(\|\mathbf{v}^{\mathcal{B}}\|)\|\mathbf{v}^{\mathcal{B}}\| + \tanh^2(\|\mathbf{v}^{\mathcal{B}}\|)\|\mathbf{x}^{\mathcal{B}}\|^2\|\mathbf{v}^{\mathcal{B}}\| + 4 \tanh(\|\mathbf{v}^{\mathcal{B}}\|)\mathbf{x}^{\mathcal{B}} \cdot \mathbf{v}^{\mathcal{B}}} \\ &= \frac{2 (v + 2xv \cos(\theta) \tanh(v) + v \tanh^2(v)) \mathbf{x}^{\mathcal{B}} + 2 (\tanh(v) - x^2 \tanh(v)) \mathbf{v}^{\mathcal{B}}}{v + x^2v + 4xv \cos(\theta) \tanh(v) + \tanh^2(v)v + x^2v \tanh^2(v)} \\ &= \frac{2 (1 + 2x \cos(\theta) \tanh(v) + \tanh^2(v))}{1 + x^2 + 4x \cos(\theta) \tanh(v) + \tanh^2(v) + x^2 \tanh^2(v)} \mathbf{x}^{\mathcal{B}} \\ &+ \frac{2 (\tanh(v) - x^2 \tanh(v))}{v (1 + x^2 + 4x \cos(\theta) \tanh(v) + \tanh^2(v) + x^2 \tanh^2(v))} \mathbf{v}^{\mathcal{B}}. \end{aligned} \quad (86)$$

The expressions in Equation (85) and Equation (86) agree. Therefore, $\mathbf{x}^{\mathcal{K}} \oplus_{\mathbf{E}} \exp_{\mathbf{o}^{\mathcal{K}}}^{\mathcal{K}}(\mathbf{b}^{\mathcal{K}})$ and $\mathbf{x}^{\mathcal{B}} \oplus_{\mathbf{M}} \exp_{\mathbf{o}^{\mathcal{B}}}^{\mathcal{B}}(\mathbf{b}^{\mathcal{B}})$ are the corresponding points in the hyperbolic space. We conclude that

$$P_{\mathbf{o}^{\mathcal{K}} \rightarrow \mathbf{x}^{\mathcal{K}}}(\mathbf{v}^{\mathcal{K}}) = \log_{\mathbf{x}^{\mathcal{K}}}^{\mathcal{K}} (\mathbf{x}^{\mathcal{K}} \oplus_{\mathbf{E}} \exp_{\mathbf{o}^{\mathcal{K}}}^{\mathcal{K}}(\mathbf{v}^{\mathcal{K}})). \quad (87)$$

■

B.6. Proof of Theorem 9

A simple application of Equation (12) yields

$$\begin{aligned} \mathbf{M}^{\otimes_{\mathbf{E}}}(\mathbf{x}^{\mathcal{K}}) &= \exp_{\mathbf{o}^{\mathcal{K}}}^{\mathcal{K}}(\mathbf{M} \log_{\mathbf{o}^{\mathcal{K}}}^{\mathcal{K}}(\mathbf{x}^{\mathcal{K}})) \\ &= \tanh \left(\frac{2\|\mathbf{M}\mathbf{x}^{\mathcal{K}}\|}{\|\mathbf{x}^{\mathcal{K}}\|} \tanh^{-1} \left(\frac{\|\mathbf{x}^{\mathcal{K}}\|}{1 + \sqrt{1 - \|\mathbf{x}^{\mathcal{K}}\|^2}} \right) \right) \frac{\mathbf{M}\mathbf{x}^{\mathcal{K}}}{\|\mathbf{M}\mathbf{x}^{\mathcal{K}}\|}. \end{aligned} \quad (88)$$

■

Appendix C. Statistics of Datasets

The following table summarizes the statistics of the datasets.

Table 3: Statistics of datasets

Dataset	Texas	Wisconsin	Chameleon	Actor	Cora	Pubmed
Nodes	183	251	2,277	7,600	2,708	19,717
Edges	280	466	31,421	26,752	5,278	44,327
Features	1,703	1,703	2,325	931	1,433	500
Class	5	5	5	5	7	3
Hyperbolicity	1.0	1.0	1.5	1.5	3.0	2.5

Appendix D. Additional Experiments

To study how training loss changes with training epochs using different models, we plot their relationships when training on some selected datasets. Figure 2 shows the result on the Wisconsin, Cora, and Pubmed datasets. We observe that the training dynamics of the HNNs are very similar.

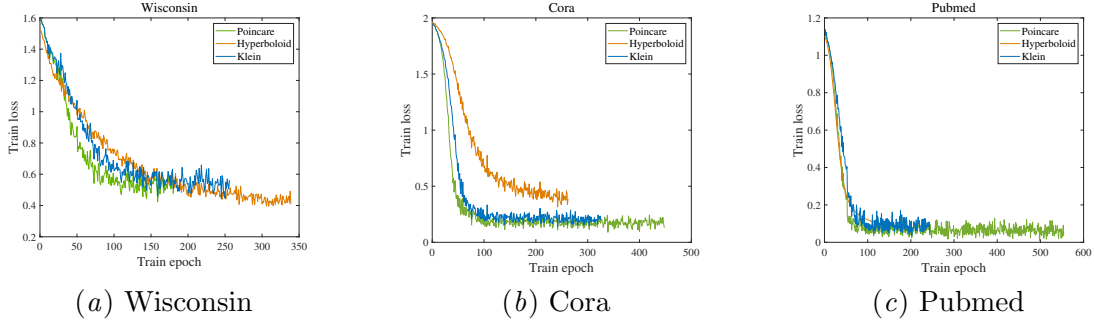


Figure 2: Change of training loss with epochs using different models.